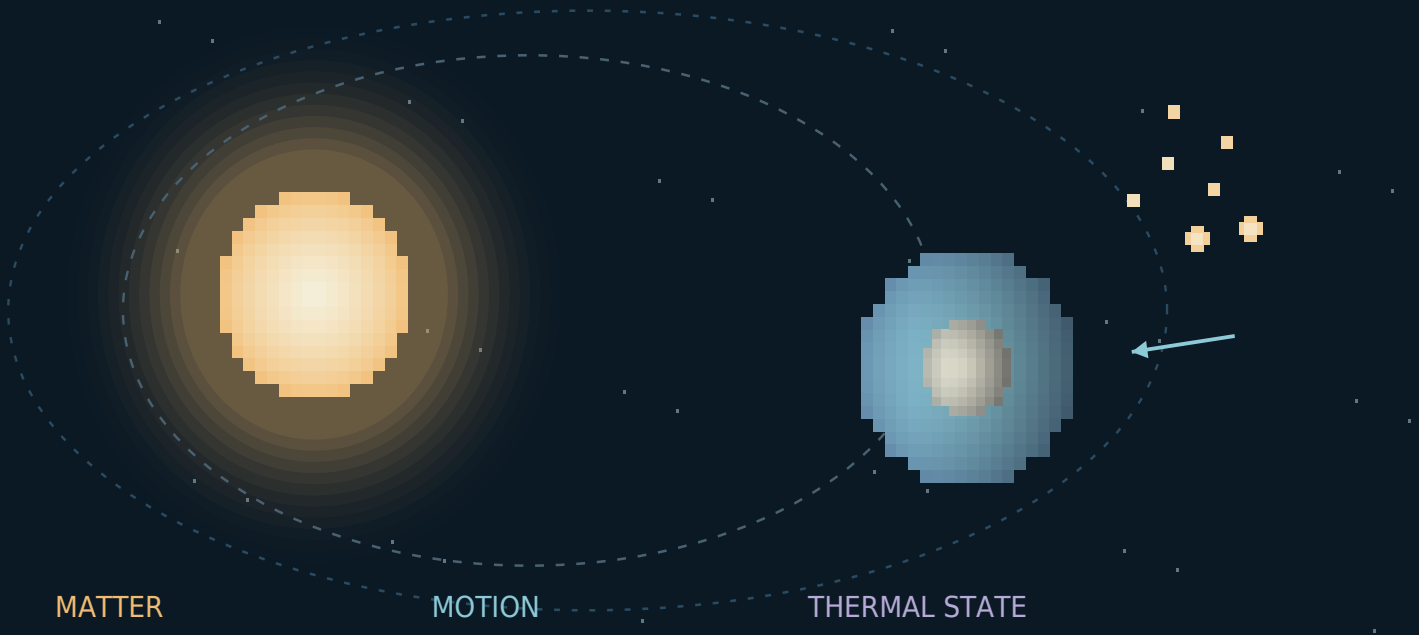
PROJECT KUJIRA

A rasterized, pixelated 2D universe



MATTER / MOTION / THERMAL STATE / TIME

An illustrated implementation atlas of planetary growth, stellar artifacts, material combinations and the mathematical rules connecting them.

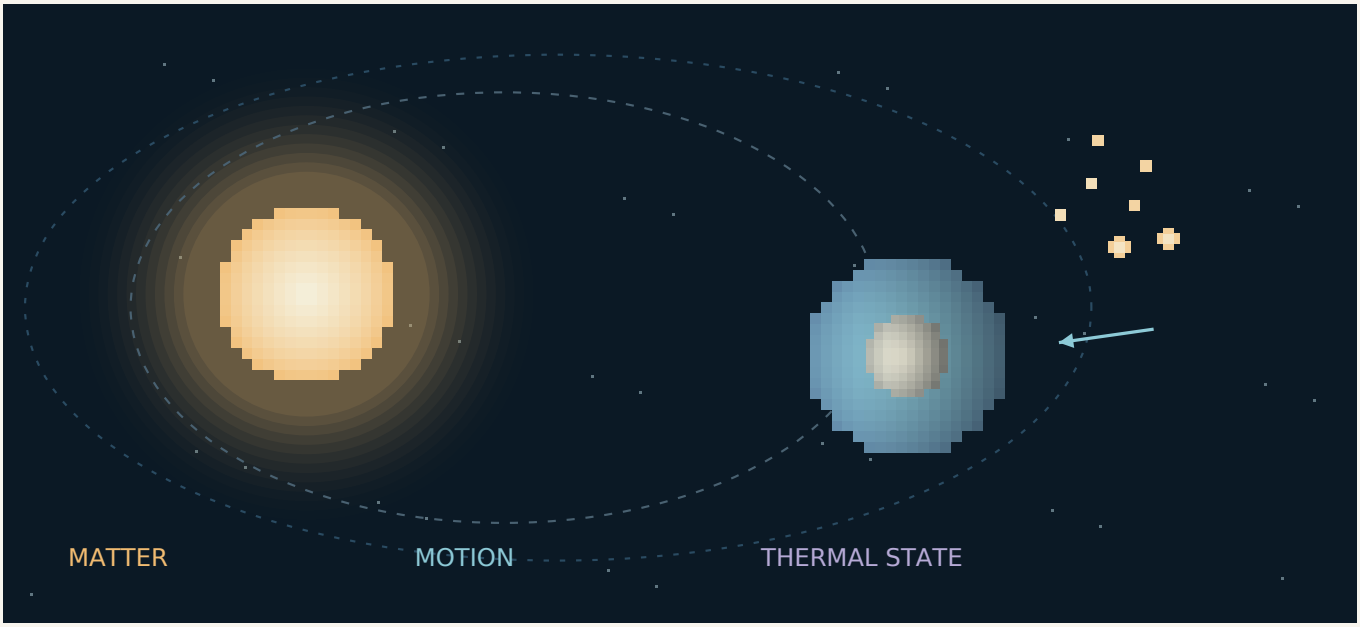
GAME 16.3.1-dev | SAVE FORMAT 16

GitHub @kujiwa-rgb / X @officialemane

7 October 2026 / Original project material copyright 2026

Preface & abstract

A development atlas of matter, motion and observation.



A conceptual universe drawn from particles, parcels, worlds and their shared records.

Abstract. Project Kujira is an open-world, rasterized, pixelated two-dimensional universe under development. It represents matter as a bounded collection of entities whose retained species, heat, momentum and angular momentum evolve through gravity, contact, capture, reaction and emission. The picture is planar, but packet masses are derived from three-dimensional reference volumes. This distinction lets a small entity budget express systems far larger than individual pixels.

Version 16.3.1-dev extends the planetary and stellar branches with real-density packet calibration, self-gravity contraction, hierarchical orbital zones, bound debris, moonlet assembly, universe time, body ages and observer-clock readouts.

This edition replaces obsolete v15 numerical examples, not the project's history. It preserves the earlier illustrated manual as an archive and separates current implementation, explanatory diagrams, reported evidence and proposed work.

The simulation is not a complete hydrodynamic or relativistic universe solver. Softened gravity, finite reservoirs, response laws, simplified stellar mass-radius relations and unresolved-collision models remain explicit approximations. The illustrations explain those choices rather than presenting them as measurements.

Authorship. Concept and project: GitHub @kujiwa-rgb; X @officialemane. Prepared from the v16.3.1 handoff, local source and project documentation, dated 7 October 2026.

Documentation edition V4. Game version 16.3.1-dev. Save format 16.

SOURCES S01-S05 / register p.52

Reading map

Read by scale, by material or by the question on your HUD.

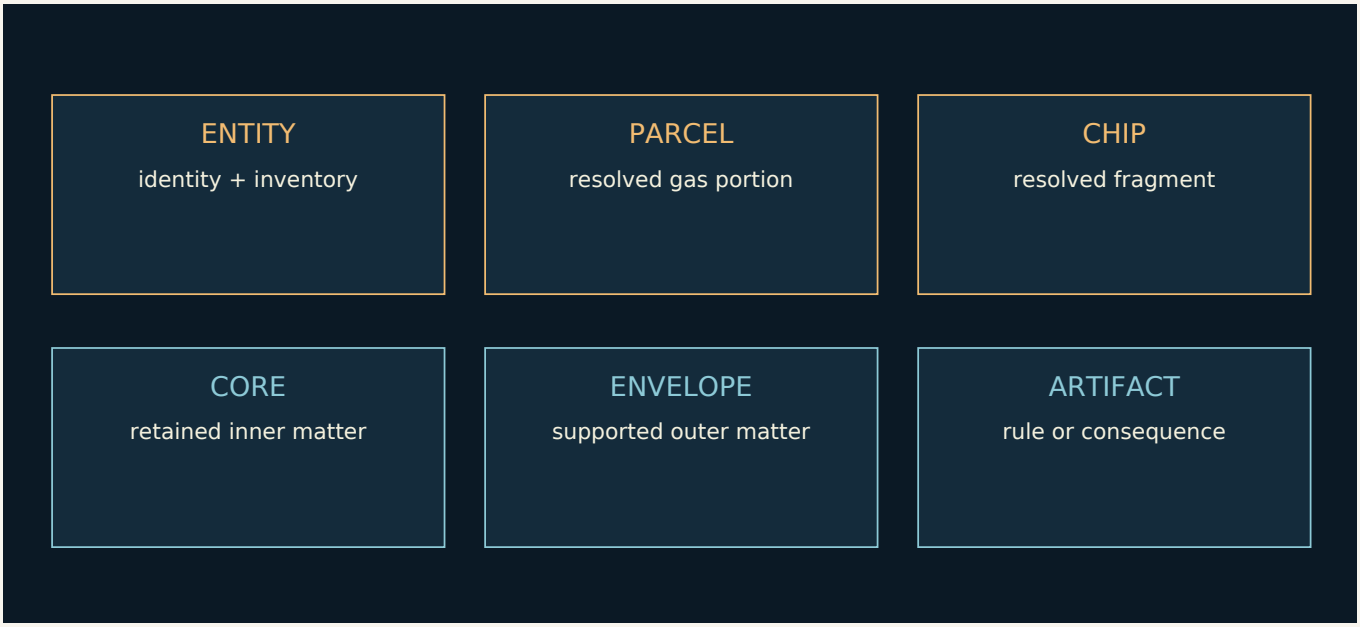
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Read in sequence, or follow the page links. Equations describe the implemented model or an explicitly labelled explanatory identity. All diagrams are original redraws; none are presented as gameplay captures.

CURRENT / CALCULATED / SCHEMATIC / REPORTED / PROPOSED

Grammar of the universe

An entity is a record of matter; an artifact is a rule or visible consequence.



The same retained inventory can produce several appearances without creating additional matter.

$$D = 2R, \qquad X_s = \frac{m_s}{M}, \qquad \sum_s X_s = 1 \tag{4.1}$$

Entity. A simulated body with identity, position, velocity, mass, species and energy. **Parcel** names a resolved portion of gas. **Chip** names a fragment, not a different chemical element. **Core** and **envelope** are regions or reservoirs within an entity.

Species means a mass inventory such as Hydrio or Irio. **Phase** describes material state or response. **Composition** is the set of retained species fractions. A hot color alone changes none of these.

Artifact describes an implemented effect: a thermal halo, stellar transition, paid wake, zone or milestone. **Seed** denotes a stellar-branch candidate, not every large gas cloud.

GAS+ records gas-majority progress. **GIANT+** records a compressed-size threshold. **GROUP** describes nearby bodies being inspected together; it is not proof of physical merger.

Throughout this book, R means radius, D means diameter, M or m means mass, and T denotes temperature unless a period is explicitly named.

Gas percentage is a mass fraction, not a percentage of the drawn radius.

SOURCES S01,S11,S13 / register p.52

From v15 to v16.3.1

Preserve the branch logic; replace obsolete scale assumptions.



A documented evolution, not a reconstruction of missing Git commits.

v15 archive. The September illustrated manual described the planetary/stellar tree, contact ledgers, heat reservoirs and a pixelated view. Its packet masses and several examples are now historical.

October 6 restoration. Hot-source glow was restored, free-cloud gravity softened over the cloud, captured gas compression reduced from eightfold to fourfold, and a 750-km atmospheric shell floor added.

v16.0. Real cold reference densities, collapse heating, new selectable scales, warp and save conversion.

v16.1. Orbit-scaled settling, periapsis-aware capture and Roche overflow. **v16.2.** Belt threshold, moonlet assembly and nested zones.

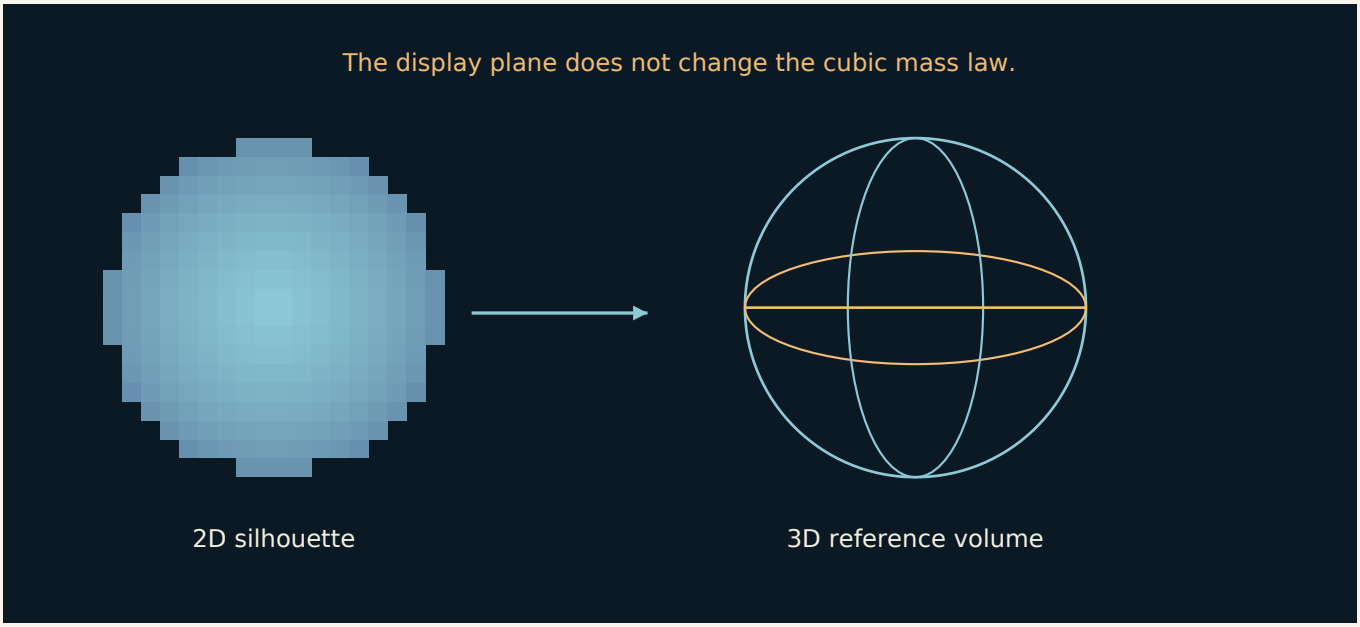
v16.3 / 16.3.1. Universe time, birth ticks, observer lag and separate moon ORBIT versus system YEAR. Save format is 16, not the document edition number.

Older examples in FILE_MAP and orbital-zone notes are explicitly superseded where their masses or save versions conflict.

SOURCES S01-S05,H01 / register p.52

A 2D window onto 3D mass

The image is planar. The density-volume calibration is spherical.



Left: a pixel footprint. Right: the volume used for packet mass. The drawing is not a voxel simulation.

$$V = \frac{4\pi R^3}{3}, \qquad m = \rho V \tag{6.1}$$

$$\frac{m(2D)}{m(D)} = 8 \quad \text{at fixed composition} \tag{6.2}$$

Every gas or condensed packet uses a reference mass computed from density times spherical volume. A diameter twice as large therefore represents eight times as much of the same material, not four times.

The simulation's positions, contact paths and displayed orbits lie in a plane. Circular silhouettes and pixel cells do not turn the mass law into a two-dimensional surface-density law.

The 375-km fragment is the base resolved size and spans one pixel at zoom 1. A fragment still carries a complete inventory and energy budget. It is not one atom and it is not a disposable visual particle.

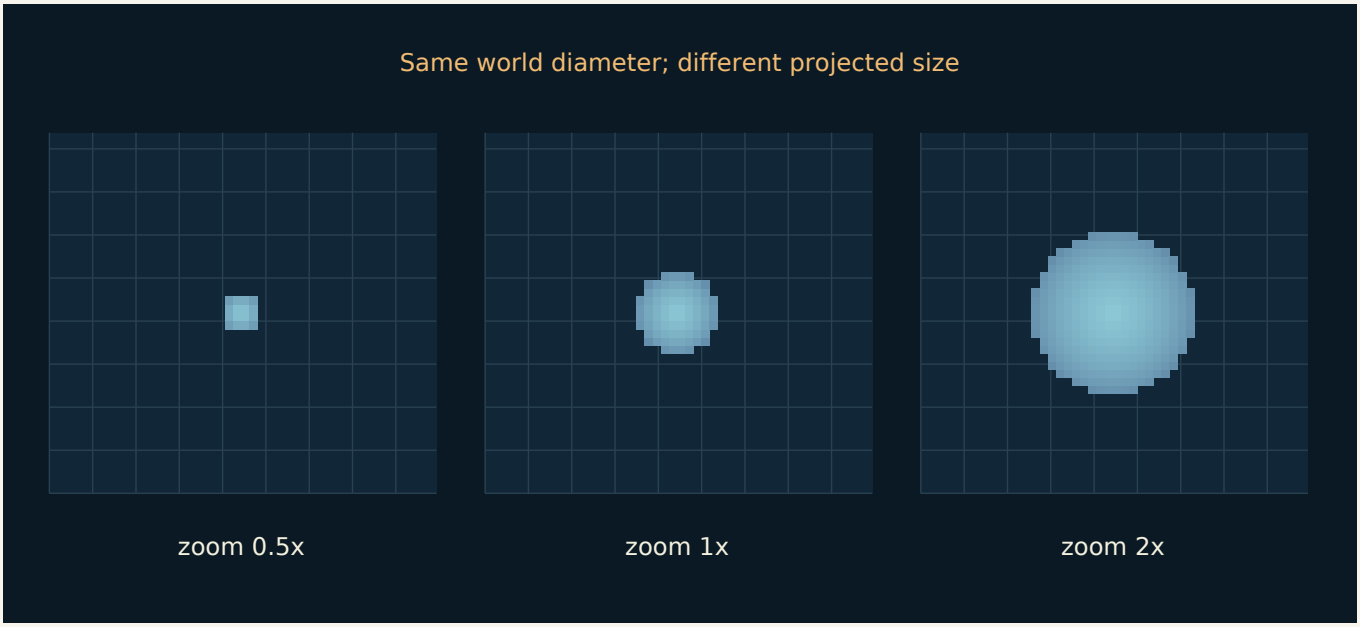
Cold reference densities calibrate packet masses. They are not a pressure-dependent equation of state evaluated continuously from local temperature. Gas-like response and the density reference are separate modeling decisions.

Pixel area is artwork; kilograms come from the reference volume.

SOURCES S02,S06 / register p.52

World mapping & zoom

One ruler connects kilometres, world coordinates and screen pixels.



Adaptive grid examples use the 1-2-5 ladder. Zoom changes projection, not retained size.

$$x_{\mathrm{px}} = x_{\mathrm{centre}} + 8z(x_{\mathrm{world}} - x_{\mathrm{camera}}) \tag{7.1}$$

$$1 \text{ world unit} = 3000 \text{ km}, \qquad D_{\mathrm{px}} = \frac{8zD_{\mathrm{km}}}{3000} \tag{7.2}$$

One world unit is 3,000 km. At zoom 1 it spans eight screen pixels. The base navigation cell remains 1,000,000 km, while adaptive display cells can descend to 10,000 km. At zoom 1 the displayed major cell is 50,000 km.

Positions are transformed relative to the camera. The screen's vertical convention is a rendering choice; it must not change the underlying vector arithmetic.

CELL and FAINT describe visible grid spacing. Their values should be read before judging the apparent distance between a star, a planet and a parcel.

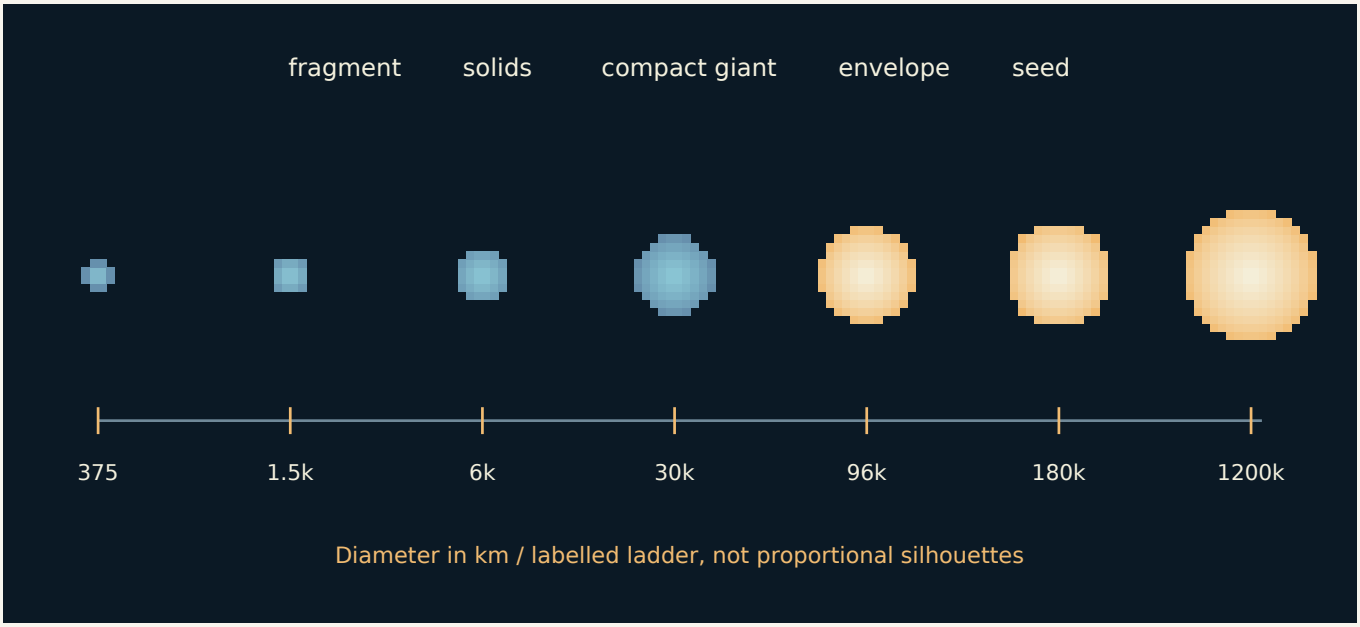
A minimum marker can keep subpixel matter discoverable. It is not an enlargement of the physical collider, the atmospheric inventory or the gravitational mass. A glow is likewise not a boundary at which objects must touch.

The scale diagram labels distances; orbital diagrams elsewhere may be schematic.

SOURCES S01,S02,S04 / register p.52

Diameter atlas

Compare sizes without confusing a cloud, a compressed world and a star.



Logarithmic diameter ruler in kilometres; glyph sizes are symbolic, not to the same linear scale.

The size ladder begins at a 375-km fragment and includes 1,500-, 3,000-, 6,000- and 12,000-km bodies. Gas presets extend through 48,000 and 96,000 km, then larger 192,000-768,000-km clouds before the stellar seed.

Solid spawning reaches 12,000 km in v16. The retained solid-growth ceiling is 20,000 km. These values are not the same as the 30,000-km compressed GIANT+ threshold.

The planetary envelope ceiling is 180,000 km. Incoming cloud eligibility, live diffuse extent and compressed equilibrium size are different decisions; no single diameter substitutes for all of them.

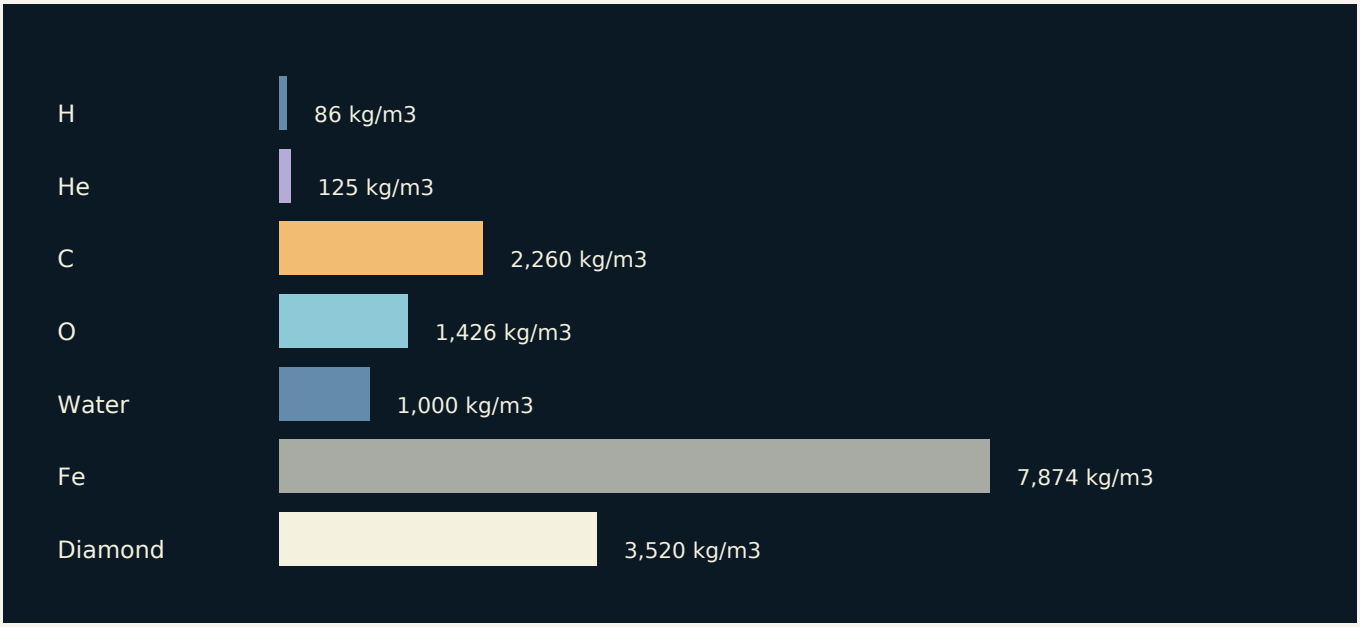
A star's displayed radius follows its stellar mass-radius branch. The approximately 1.2-million-km stellar seed is a preset context, not a universal final stellar diameter.

A large selectable free cloud is not automatically eligible to become a planet.

SOURCES S01,S02,S11 / register p.52

Material density atlas

Real reference density replaces the old packet-mass calibration.



Reference densities in kg/m³. Bars use a logarithmic scale to retain visibility of H and He.

$$m_s(D) = \rho_s \frac{\pi}{6} (1000 D_{\mathrm{km}})^3 \tag{9.1}$$

Hydrio uses 86 kg/m³, Helio 125, Carb 2,260, Oxio 1,426, water 1,000, Irio 7,874 and Diamond 3,520. These are the current source constants.

Hydrio and Oxio use cold condensed reference states, Helio a liquid reference, Carb graphite, Irio iron and water a liquid calibration. The names remain Kujira's material vocabulary.

At the same diameter, mass ratios equal density ratios. At different diameters, the cubic diameter factor matters just as much. Two equally bright sprites need not have equal mass.

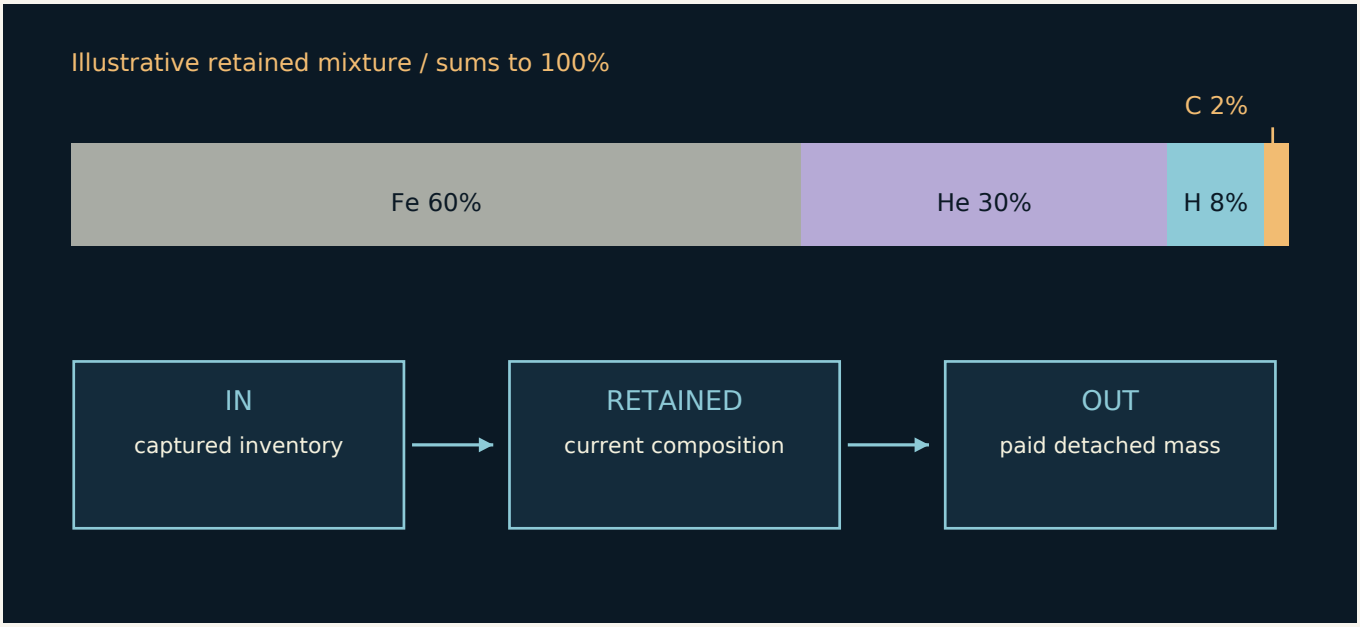
A 375-km Irio fragment is about 2.17 × 10²⁰ kg. A 1,500-km Irio body is about 1.39 × 10²² kg. A 3,000-km Hydrio packet is about 1.22 × 10²¹ kg.

Reference-state calibration is not a full thermodynamic density model.

SOURCES S02,S06 / register p.52

Inventory & percentages

The HUD describes retained kilograms, not colors or nearby ownership.



One illustrative ledger: species add to total mass; display regions are derived from that ledger.

$$M = \sum_s m_s, \qquad \mathrm{HUD}_s = 100\frac{m_s}{M}\ \% \tag{10.1}$$

$$m_s' = m_s + \Delta m_{s,\,\mathrm{in}} - \Delta m_{s,\,\mathrm{out}} + \Delta m_{s,\,\mathrm{reaction}} \tag{10.2}$$

The authoritative composition is the mass of each retained species. Every capture, split, chemical change and stellar process must update that inventory. A layer may become opaque without adding a kilogram.

A source losing a mixed parcel loses the parcel's actual species. Reaccretion adds those same inventories back. Reaction products appear only when the corresponding conversion has occurred.

An interaction group can include detached gas and a solid. GROUP MASS and GROUP DIAM describe that observation, not necessarily one merged atmosphere. The distinction matters when a large cloud is still settling around a small core.

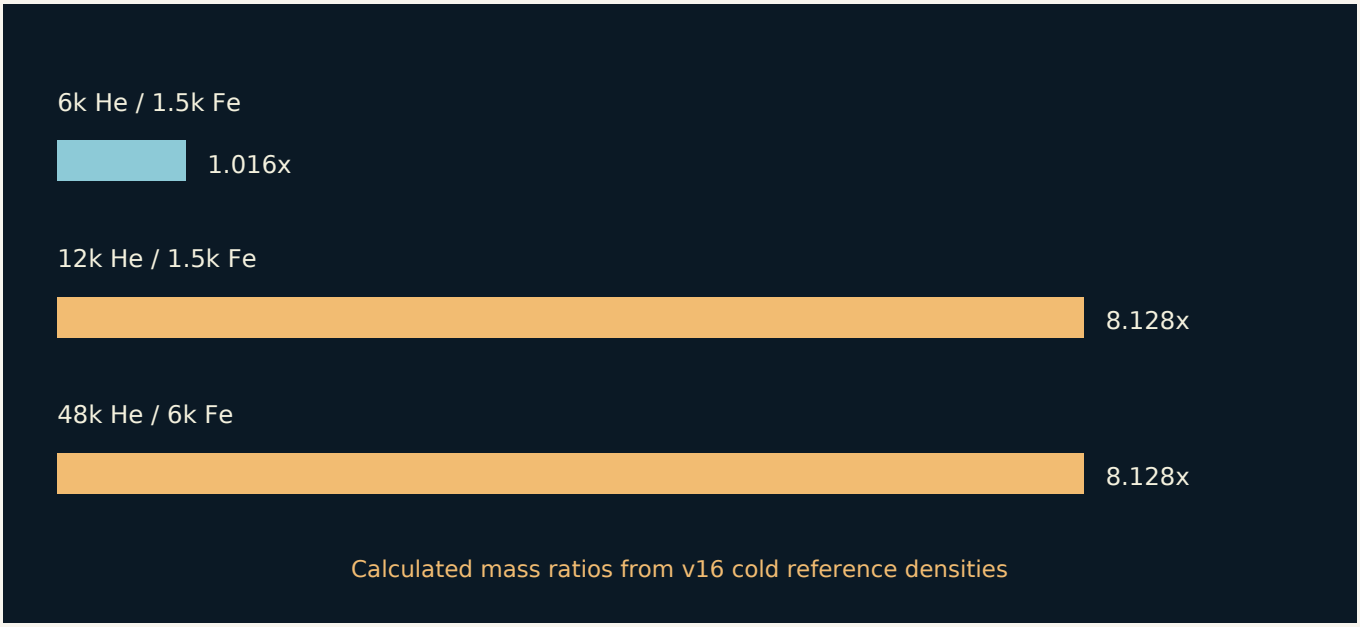
Trace fractions can be numerically real while their geometric layer is below the current pixel scale. Enlarging a trace shell to look important would misrepresent its mass share.

Stored water is compound inventory; do not double-count its constituents as additional free H and O.

SOURCES S01,S11,S13 / register p.52

Mass ratios, not diameter guesses

The v16 density change invalidates several earlier verbal recipes.



Calculated from the current reference densities; these are inventory comparisons, not completed captures.

$$\frac{m_{\mathrm{He}}}{m_{\mathrm{Fe}}} = \frac{125}{7874}\left(\frac{D_{\mathrm{He}}}{D_{\mathrm{Fe}}}\right)^3 \tag{11.1}$$

A 6,000-km Helio cloud has about 1.02 times the mass of 1,500-km Irio. A 12,000-km Helio cloud has about 8.13 times that Irio mass. The earlier 'half as heavy' and 'four times' comparisons belonged to the older calibration.

The same 8:1 diameter ratio occurs for 48,000-km Helio and 6,000-km Irio, so their initial mass ratio is again about 8.13.

Ratios do not promise instant capture. A large incoming cloud can cross, shear, leave a coating, lose pieces and return. Retention still depends on eligibility, exposure, geometry, motion and the energy ledger.

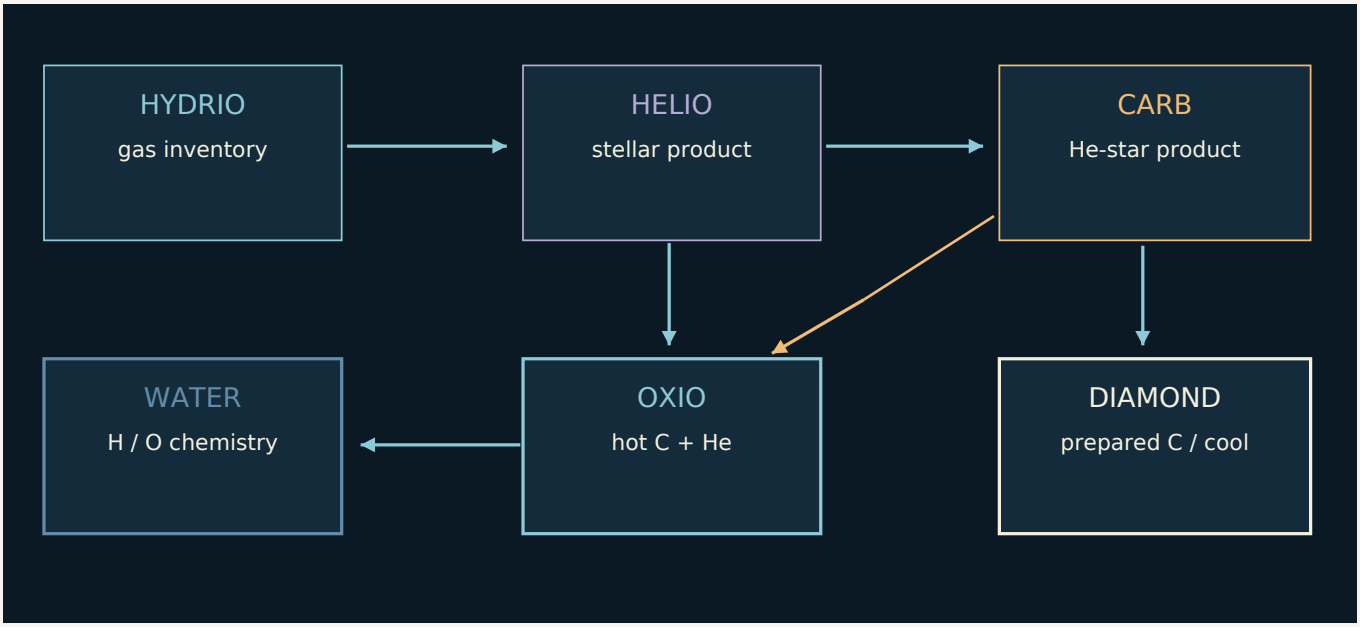
After full retention, gas fraction is $m_g / (m_g + m_s)$. Compression changes envelope volume, not this mass fraction. The worked assembly on page 25 follows those quantities separately.

Calculated examples are not a substitute for measuring how much gas actually survives capture.

SOURCES S02,S06 / register p.52

The artifact branching map

Mixing, chemical conversion, nuclear conversion and cooling are different edges.



Solid arrows: documented material routes. Dashed edges: conditions/history, never material created from nothing.

Hydrio stellar processing produces Helio. The Helio stellar branch produces Carb; hot, dense Carb-Helio exposure can produce Oxio. Merely touching cool parcels is not nuclear conversion.

Hydrio and Oxio can contribute to water under the chemical rule. Water changes phase through its enthalpy model. Hot processing can return constituents, with their masses accounted.

Prepared surviving Carb may follow the Diamond cooling-history artifact. Diamond is a separate retained solid inventory, not an extra copy of its source carbon. Its preparation score cannot create material.

Irio supplies the condensed planetary branch. Captured gas and a settled core can form a gaseous planet, while the stellar branch requires the appropriate gas state and stellar thresholds. Cored planets do not ignite as ordinary pure-gas stars.

The branching map is not a claim that all named transformations are realistic at every temperature or pressure.

SOURCES S01,S05,S11,S12,H01 / register p.52

Units & dimensional grammar

A number needs a unit, a reference frame and a meaning.

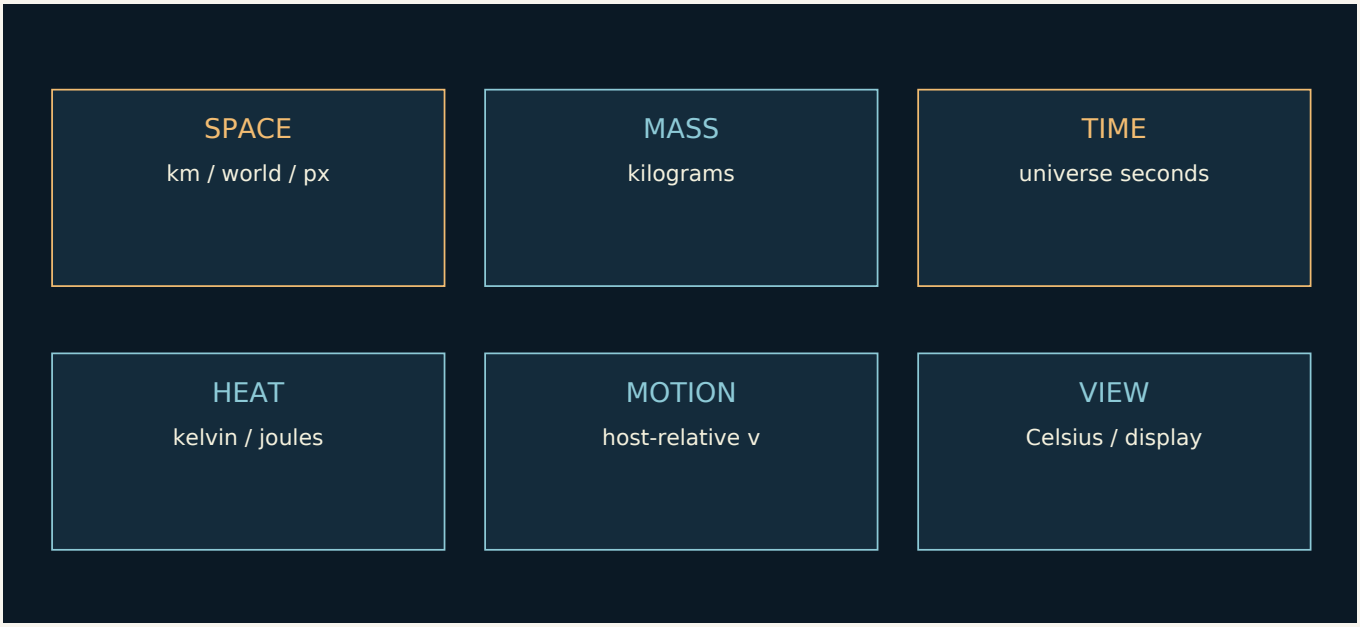


FIG. 13 / Unit bridges used throughout the model; conversions are exact where stated.

$$T_{\mathrm{K}} = T_{^\circ\mathrm{C}} + 273.15, \qquad 1\ \mathrm{km} = 10^{3}\ \mathrm{m} \tag{13.1}$$

$$G = 6.67430 \times 10^{-11}\ \mathrm{m^{3}\,kg^{-1}\,s^{-2}} \tag{13.2}$$

Internal space. A world unit represents 3,000 km. Rendering projects that length to eight pixels at zoom one. Simulation calculations must convert consistently before applying the SI gravitational constant. The pixel grid is not a new unit of mass or a physical minimum distance.

Temperature. Energy and radiation use kelvin. HUD temperatures are displayed in Celsius. A cold-space reservoir of 10 K therefore appears near -263 degrees Celsius, not zero kelvin.

Mass and time. Kilograms, seconds and joules connect the material, orbital and thermal ledgers. A Julian year is 31,557,600 seconds. The displayed universe clock advances simulation time, whereas PLAY measures real session time.

Notation. Capital M usually denotes a host, lowercase m a parcel, R a radius and D a diameter. Vector v is velocity; its magnitude alone is speed. Relative velocity must subtract the host's motion before interpreting capture or escape.

MODEL RULE / Display units can change without changing the underlying physical quantity.

SOURCES S02,S04,S06,S09 / register p.52

Gravity with a finite centre

One softened interaction must serve force, potential and escape.

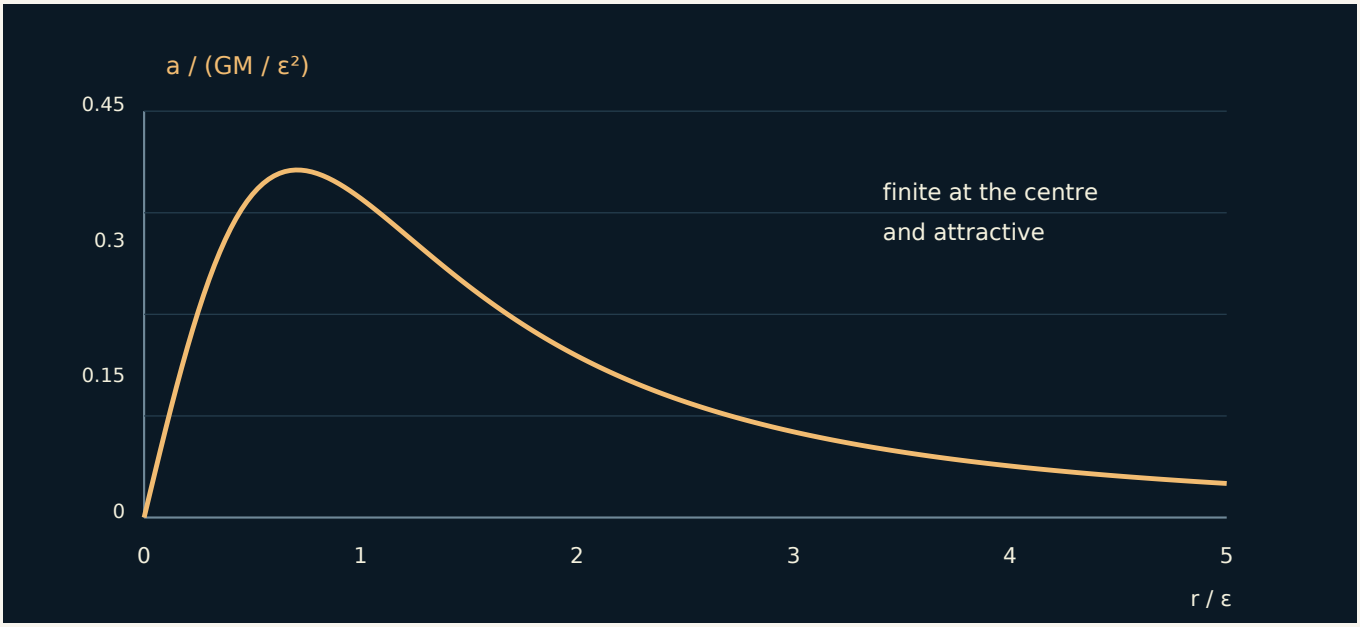


FIG. 14 / Calculated Plummer force profile against radius, normalized by the softening length.

$$\mathbf{a}_i = \sum_{j \neq i} \frac{G m_j (\mathbf{r}_j - \mathbf{r}_i)}{(r_{ij}^2 + \varepsilon_{ij}^2)^{3/2}} \tag{14.1}$$

$$U_{ij} = -\frac{G m_i m_j}{\sqrt{r_{ij}^2 + \varepsilon_{ij}^2}} \tag{14.2}$$

Pair attraction. The gravitational acceleration points toward the other body's centre. Plummer softening replaces the divergent point-mass centre with a finite potential. It is a numerical and geometric approximation, not a new repulsive force or an invisible solid wall.

Shared kernel. Force integration, escape checks and potential-energy accounting must agree on the softening length. Otherwise a collision can look bound to one subsystem and unbound to another.

Scale choices. The global reference softening is 13,889 km. Planetary pairs can use a smaller length tied to the larger body's radius. Planet/free-cloud pairs also account for the cloud's extent.

Cloud centre. A softening of two-thirds of a uniform cloud's radius matches that sphere's central potential depth. This does not reproduce its complete interior force curve. The distinction matters when interpreting a solid settling within gas.

INTERPRETATION / Gravity can support an orbit; dissipation, not attraction alone, removes orbital energy.

SOURCES S02,S06,S08 / register p.52

Momentum, orbit & spin

A packet may exchange motion, but cannot receive a free kick.

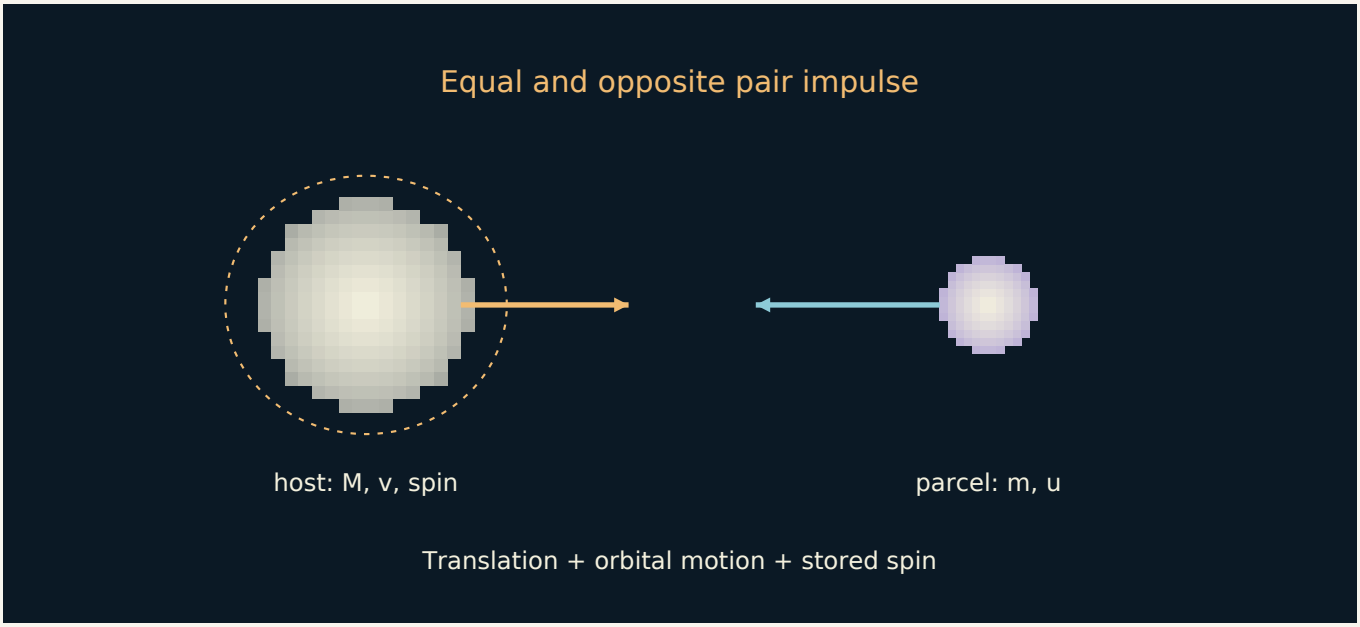


FIG. 15 / Translation, orbital angular momentum and stored spin are distinct accounts.

$$\mathbf{P} = \sum_i m_i \mathbf{v}_i, \qquad L_z = \sum_i (\mathbf{r}_i \times m_i \mathbf{v}_i)_z + \sum_i S_i \tag{15.1}$$

$$I_{\text{sphere}} = \tfrac{2}{5} M R^2, \qquad K_{\text{spin}} = \frac{S^2}{2I} \tag{15.2}$$

Translation. The sum of mass times velocity defines linear momentum. An internal pair impulse must be equal and opposite. A small parcel can change speed substantially while the much heavier host receives a subtle recoil.

Rotation. Angular momentum includes motion about the chosen origin and each body's stored spin. A merge can move orbital angular momentum into spin, rather than silently deleting it.

Energy is separate. Conserving momentum does not automatically conserve kinetic energy. Drag can reduce relative kinetic energy, but the removed energy must appear as heat or another explicitly modeled reservoir.

Changing size. Contraction changes the moment of inertia. For an ideal uniform sphere it scales with mass times radius squared. Spin therefore becomes more expensive as the body contracts, and the contraction budget must account for that cost.

REFERENCE / The sphere formula explains scaling; composite-body bookkeeping follows its actual model inertia.

SOURCES S01,S03,S07 / register p.52

The contact ledger

Material transfer, recoil and heating are one transaction.

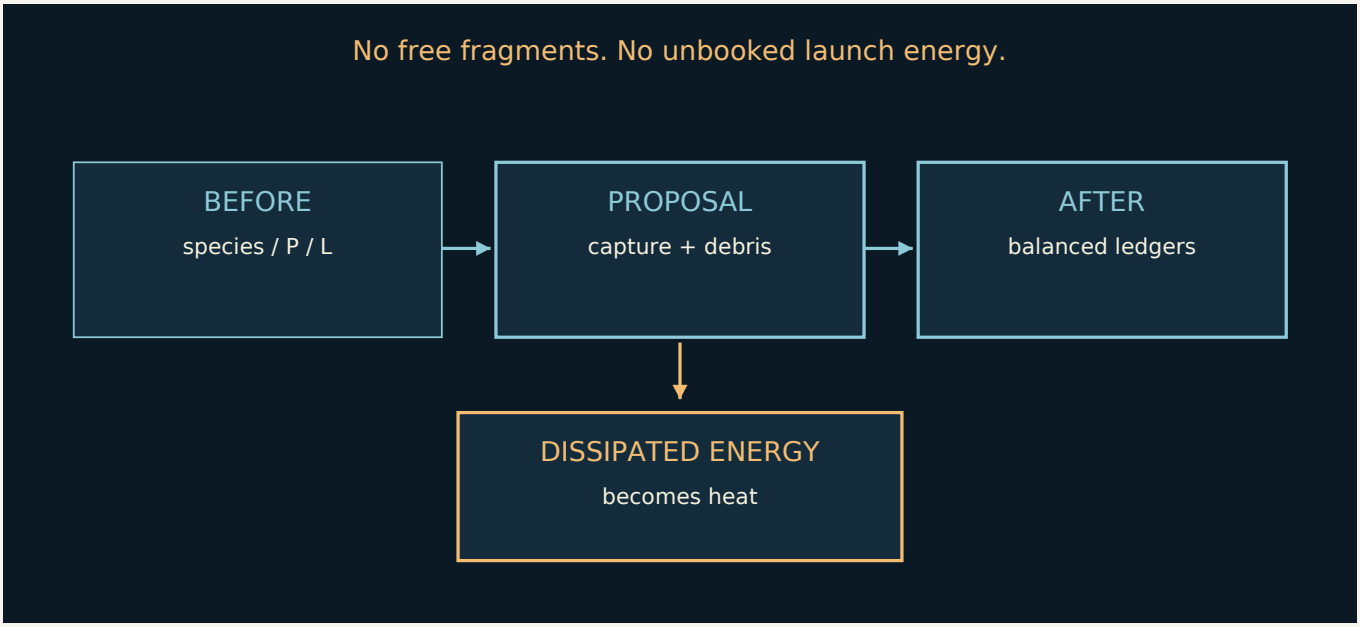


FIG. 16 / A contact proposal is accepted only with a consistent material and energy ledger.

$$\mu = \frac{m_1 m_2}{m_1 + m_2}, \qquad K_{\mathrm{rel}} = \tfrac{1}{2}\mu |\mathbf{v}_1 - \mathbf{v}_2|^2 \tag{16.1}$$

$$M_{\mathrm{before}} = M_{\mathrm{remnant}} + M_{\mathrm{cloud}} + \sum_k m_{\mathrm{debris},\,k} \tag{16.2}$$

Before contact. Record the participants' species inventories, positions, velocities, thermal reservoirs and spin. Distinguish gas flow, solid impact, stellar absorption and chemical or nuclear conversion before choosing a response.

During transfer. Material removed from one participant appears in the recipient or in explicitly created debris. A fragment is not a decorative particle carrying invented mass.

After contact. The remnant, surviving cloud and detached parcels together must balance the incoming inventory and momentum. Dissipated motion becomes heat. Paid ejection and radiation are explicit outgoing terms.

Why relative speed? Two objects sharing a fast stellar orbit may touch gently. Using their absolute world speeds as impact energy would create a violent collision from what is actually a small local velocity difference.

BOUNDARY / Nuclear mass defects are separately booked; the ordinary contact identity has no reaction term.

SOURCES S01,S05,S11 / register p.52

Gas flow & broken wakes

Passing through is a residence-time interaction, not a solid bounce.

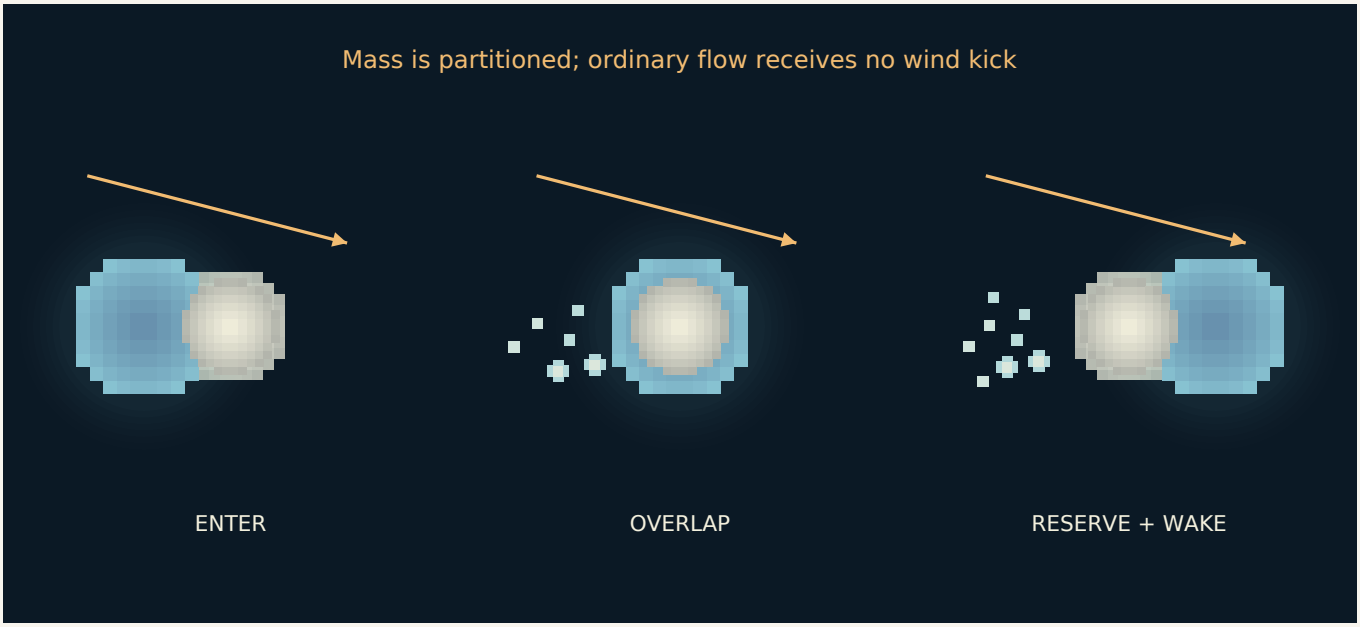


FIG. 17 / Schematic oblique transit: capture, surviving cloud and a bounded, mass-paid wake.

$$\mathbf{v}_{\mathrm{rel}}' = \lambda \mathbf{v}_{\mathrm{rel}}, \quad 0 \le \lambda \le 1 \tag{17.1}$$

$$Q_{\mathrm{drag}} = \tfrac{1}{2}\mu(1-\lambda^2)|\mathbf{v}_{\mathrm{rel}}|^2 \tag{17.2}$$

Gas is not a rigid surface. A cloud can overlap a solid while drag, heating and partial capture occur. Brief transit leaves a reserve; sustained overlap and settling transfer more. A fast arrival should not be replaced by unconditional whole-cloud merging.

Breakup. Detached parcels represent real shares of the incoming gas. Unequal parcel sizes and a spread wake communicate fluid-like separation, without requiring an unbounded shower of 375-km entities.

Local flow. Ordinary impact gas inherits the surrounding motion rather than receiving a stellar-wind launch kick. Pair damping converts relative motion to heat. The model protects suitable already-bound interactions through conservative impulses and launch preflight.

Limits. This is a finite parcel approximation, not a solved Navier-Stokes fluid. Stellar gravity, subsequent encounters and genuine energetic events can still alter an orbit. No local collision rule guarantees eternal binding throughout an arbitrary universe.

EQUATION SCOPE / This pair-damping identity explains the energy ledger; contact-dependent rates remain calibrated.

SOURCES S01,S05,S11 / register p.52

Solid impact & paid debris

A chip is removed material, not an effect painted on top.

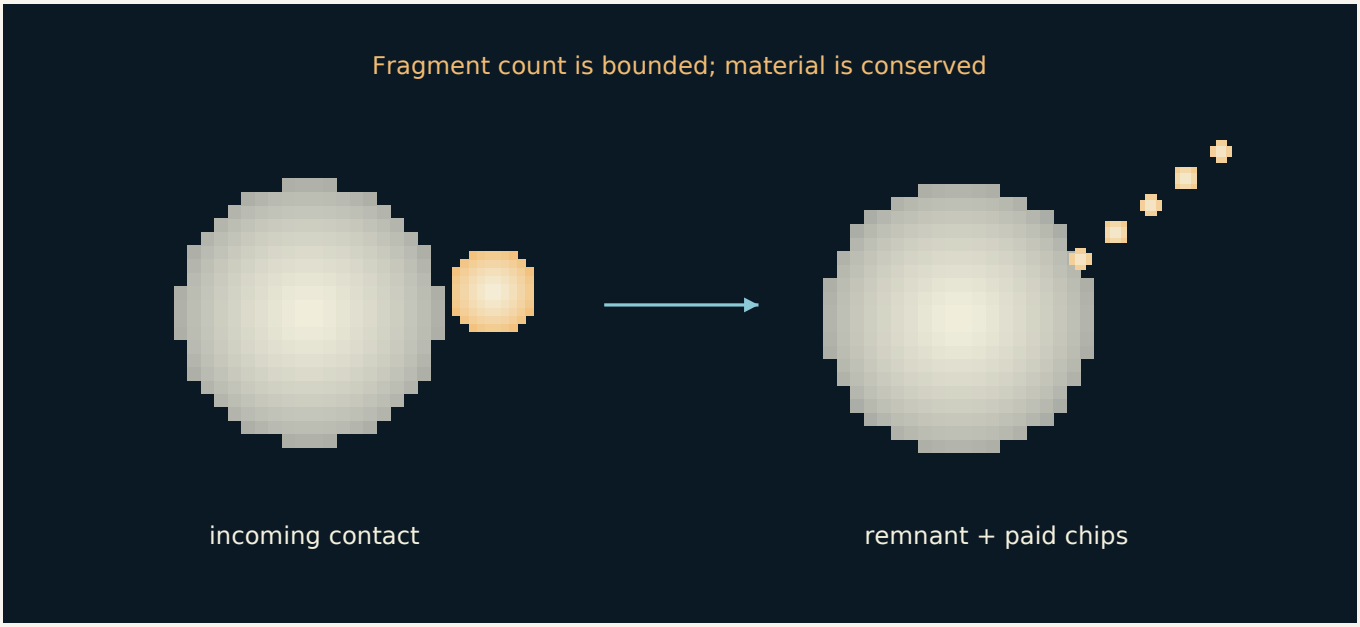


FIG. 18 / Solid contact redistributes mass between the remnant and eligible recoil debris.

$$m_{\text{chip}} = \rho_{\text{chip}} \frac{\pi D_{\text{chip}}^3}{6} \tag{18.1}$$

$$Q_{\text{available}} \geq \Delta K + \Delta U_{\text{gravity}} + \Delta K_{\text{spin}} \tag{18.2}$$

Actual contact. Solids do not acquire gas-style pass-through merely because both branches share collision detection. Physical solid contact resolves with material-specific merging, rebound or debris, while swept detection protects fast arrivals from tunnelling.

Variable outcome. Impacting mass, relative speed, geometry and available energy govern the outcome. A bounded fragment count is a performance limit, not a claim that every collision ejects the same number of pieces.

Smallest represented piece. The reference chip diameter is 375 km. A chip's mass follows its material density and volume. If an event cannot pay for its launch or satisfy the relevant orbital constraints, the proposed ejection must not create free matter.

Different destinations. A large eligible host can populate a local belt or moon-building zone. Below that threshold, debris is biased toward a compact returning trajectory rather than a permanent miniature ring system.

UNITS / Use metres in the chip-volume expression when density is in kilograms per cubic metre.

SOURCES S01,S02,S03,S08 / register p.52

Thermal reservoirs

One body can have a hot centre and cooler exposed gas.

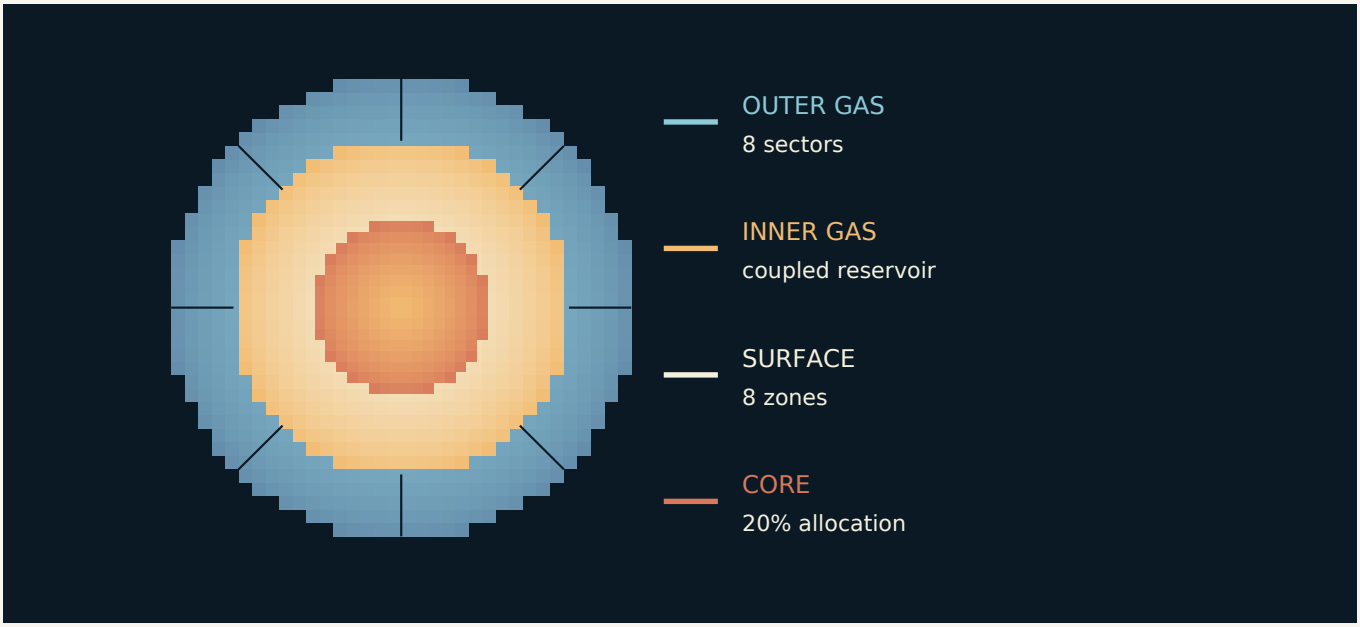


FIG. 19 / Schematic reservoirs: core, segmented surface, inner gas and outer sectors.

$$\Delta U = mc\,\Delta T, \qquad Q_{\text{phase}} = mL_{\text{phase}} \tag{19.1}$$

Structured temperature. A planetary body stores more than one thermal state. The current model uses a 20% solid-core allocation, eight surface zones, an inner gas reservoir and eight outer gas sectors. This permits directional heating and cooling.

Energy exchange. Collisions, drag and contraction can supply heat. Conduction and reservoir coupling redistribute it. Surface emission removes it. A body-wide average is useful for a group HUD but must not overwrite the component states.

Material response. The solid reference heat capacity is 450 joules per kilogram-kelvin. Its melting reference is 1,811 K with 247,000 joules per kilogram of latent heat. These are model parameters, not a full phase diagram.

Appearance. Surface-to-gas dissipation should let a hot solid warm its envelope without forcing every layer to the identical colour. Conversely, a hot core alone is not permission to display every outer parcel as white-hot.

MODEL LIMIT / Fixed reservoir fractions and coupling rates are reduced thermal mechanics, not resolved convection.

SOURCES S01,S11,S14 / register p.52

Radiation & cooling

Illumination supplies a thermal floor; empty space is not a heat sink at zero kelvin.

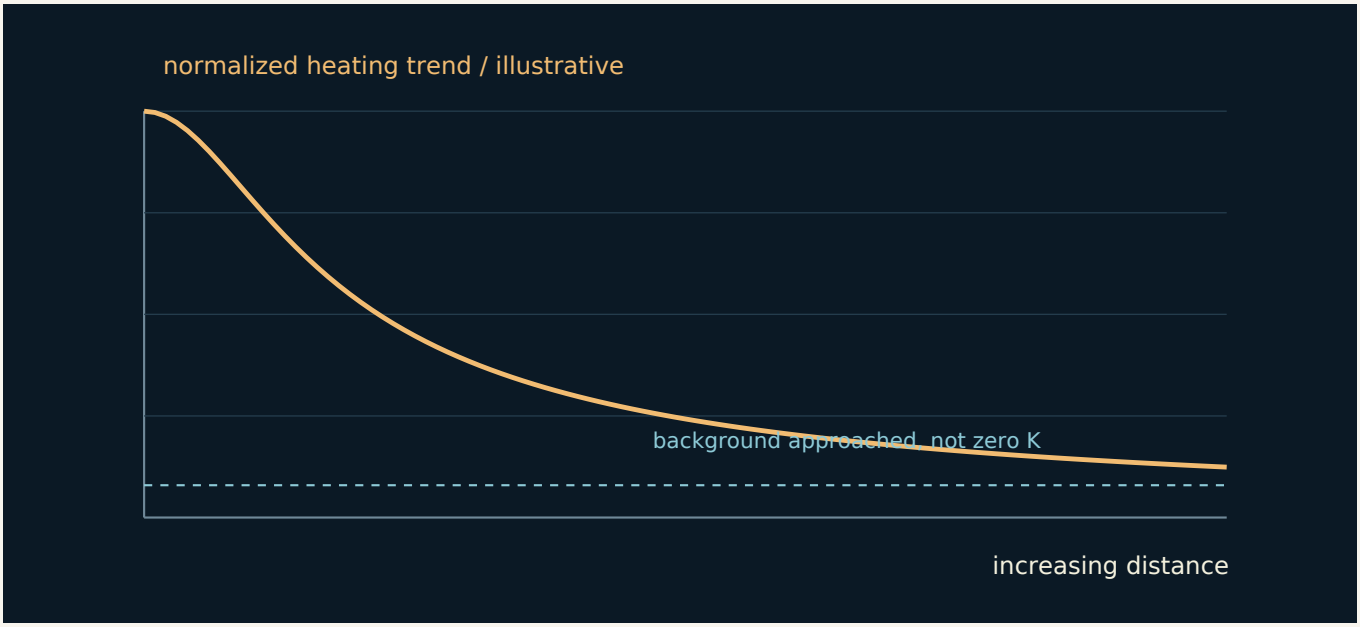


FIG. 20 / Illustrative distance-dependent heating and relaxation toward local equilibrium.

$$P_{\mathrm{net}} = \varepsilon \sigma A (T^4 - T_{\mathrm{bg}}^4) \tag{20.1}$$

$$\sigma = 5.670374419 \times 10^{-8}\ \mathrm{W\,m^{-2}\,K^{-4}} \tag{20.2}$$

External heating. A stellar source establishes an outward heating field. The local receiving state depends on source strength, separation, absorption and the object's own thermal evolution. Cooling approaches the ambient balance rather than always returning to the spawn temperature.

Cold space. The model background is 10 K. The radiation module retains a thermal-clock scale of ten million, a declared acceleration proxy. That factor should not be confused with the universe time-warp control.

Radiative bookkeeping. The Stefan-Boltzmann relationship provides the fourth-power temperature dependence. Gas emissivity and absorptivity use a reference 0.10, while planetary solid emissivity uses 0.45.

Field versus halo. A visible glow is an explanatory rendering, not a literal map of absorbed power. The game applies filtered stellar source behaviour and calibrated distance response; the equations below are the radiative foundation, not an assertion that every rendered pixel is a radiometer.

DESIGN CONTRACT / Mass and temperature affect the source; a larger halo alone must not create extra energy.

SOURCES S01,S11,S14 / register p.52

Luminosity, colour & shade

Preserve visible hot gas without turning every planet into a star.

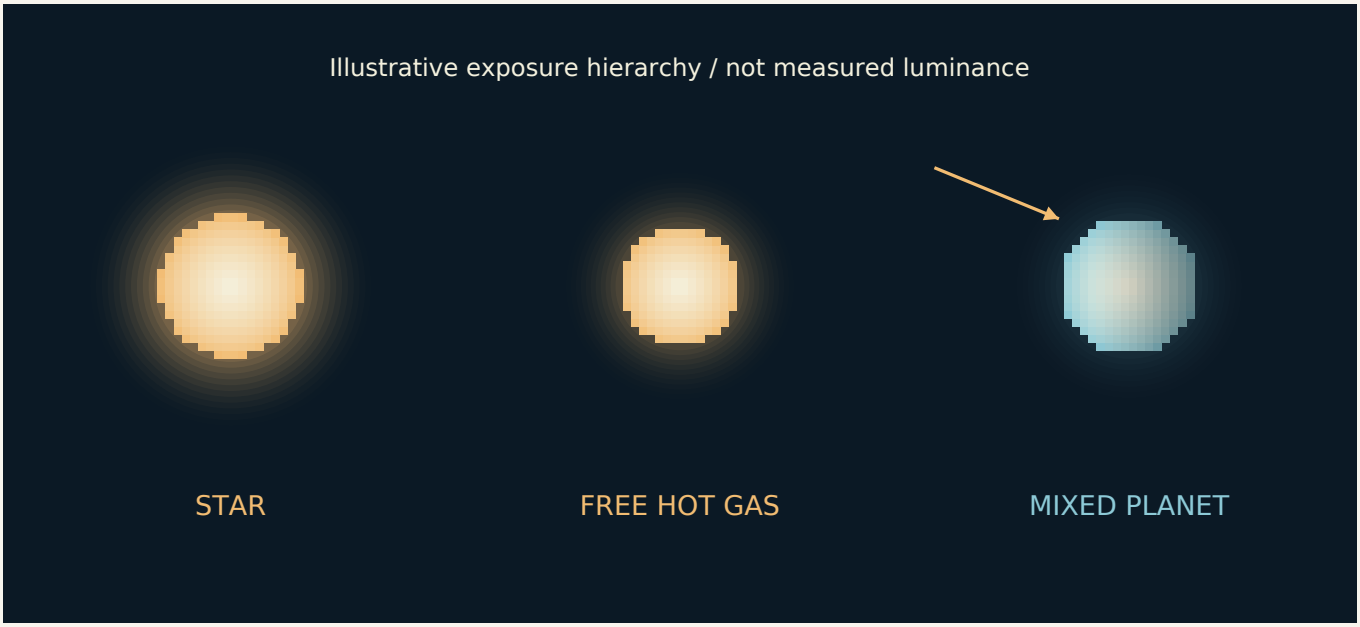


FIG. 21 / Conceptual comparison of a stellar source, free hot gas and a shaded mixed planet.

Restored visibility. The v16.3.1 handoff records restoration of the pre-October-3 hot-source field. Pure gas uses the restored luminous behaviour with its gain limiter; mixed solid-gas bodies use a more restrained exposure. These are render controls, not revised temperatures.

Direction. A planetary atmosphere shares the stellar-facing light direction with its solid host. Its night side can dim without hiding the object completely or making body spin rotate the external light source.

Temperature palette. Heat colour is not an element identity. A warm brown tint begins earlier than the older high-temperature-only palette, while the hot spectrum remains continuous. Layer composition and thermal shading are separate visual channels.

Cold envelopes. Below roughly 90 degrees Celsius, composition tint can keep retained gas legible without a hot-source halo. Bare rock does not receive an invented cold-gas atmosphere. All world-space radii continue to project through the same zoom mapping.

VISUAL STANDARD / Pixel diagrams in this atlas explain the design; they are not newly captured gameplay renders.

SOURCES S01,S05,S14 / register p.52

Composition becomes layers

The visible shell must be supported by retained inventory.

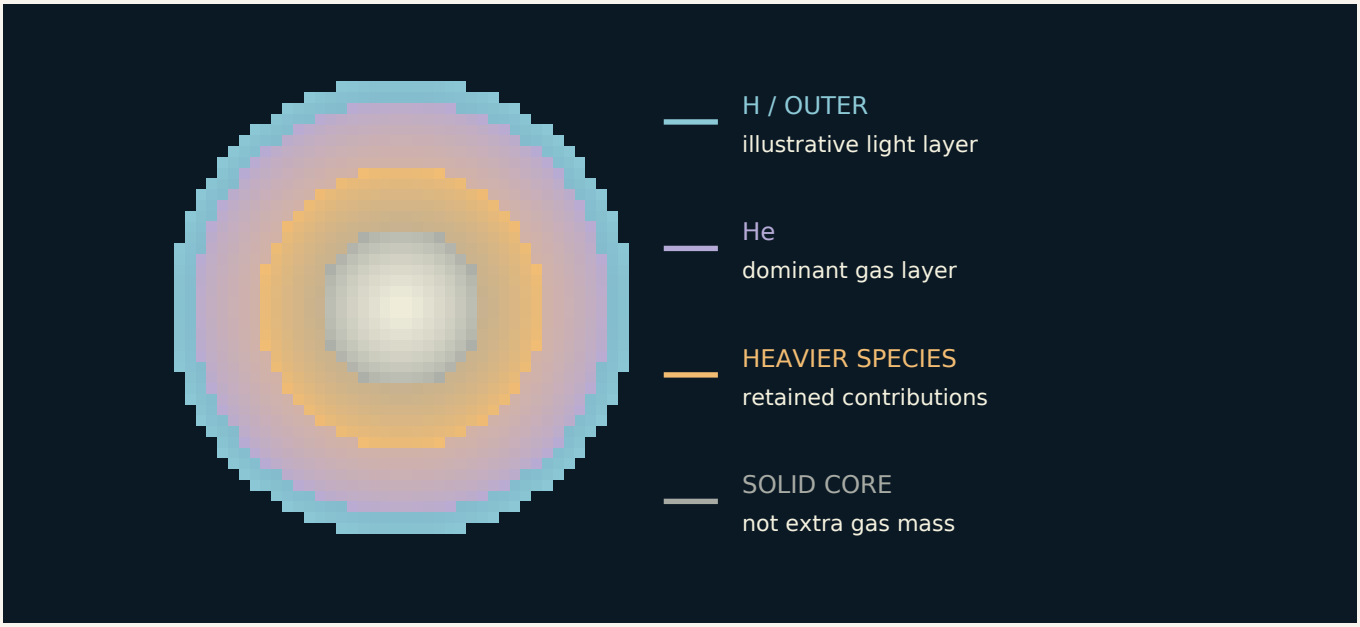


FIG. 22 / Illustrative mass-based layers; trace material remains thin, not a giant new atmosphere.

$$V_{\mathrm{shell}} = \sum_s \frac{m_s}{\rho_s C_s}, \qquad R_{\mathrm{out}}^3 = R_{\mathrm{core}}^3 + \frac{3V_{\mathrm{shell}}}{4\pi} \tag{22.1}$$

Universal accounting. Each species contributes according to its retained mass and the relevant material-volume conversion. A trace component cannot claim the same geometric thickness as a dominant gas simply because both are present.

Order and heat. The rendered envelope separates lighter and heavier contributions, while thermal coupling gives a gradient across the shell. Hydrio should be identifiable near the outside where the ordered composition model puts it.

Visibility is not extra mass. Pixel sampling can erase thin shells at low resolution. A legibility treatment may preserve a trace signal, but it must not enlarge the physical collision radius or invent inventory.

Grouped bodies. A combined interaction HUD can include a nearby cloud that is not yet retained. Its percentage must not be mistaken for a fully compressed shell. The distinction explains why BODIES greater than one is not proof of a completed giant transformation.

SCOPE / C is a prescribed compression factor; this volume sum is not a pressure-resolved equation of state.

SOURCES S01,S10,S11 / register p.52

Two planetary checkpoints

GAS+ and GIANT+ describe different achievements.

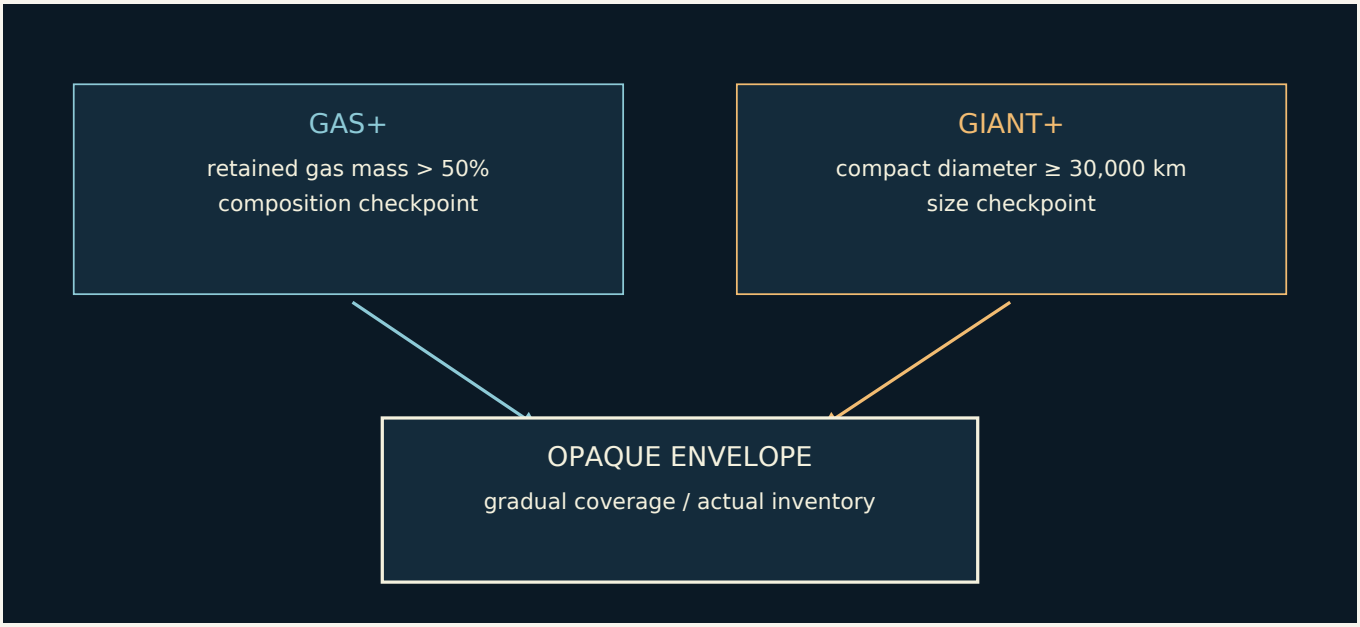


FIG. 23 / Independent composition and compressed-size gates, followed by a gradual visual response.

$$f_{\text{gas}} = \frac{M_{\text{gas}}}{M_{\text{gas}} + M_{\text{solid}}} > 0.5 \tag{23.1}$$

$$D_{\text{compact}} \geq 30{,}000 \text{ km} \tag{23.2}$$

GAS+. More than half of the retained mass is gas. This is a composition milestone, not a diameter test and not stellar ignition. A very extended but very light cloud may still carry less mass than a compact solid.

GIANT+. The compressed inventory reaches the 30,000-km diameter threshold. The reference is the finished compact state rather than the temporary, broad pre-compression envelope.

Checkpoint memory. The branch records attained milestones. Rendering still has to follow current material and coverage, so a remembered checkpoint does not justify an opaque shell after its gas has been stripped away.

Assembly safeguards. A large eligible cloud needs a sufficiently embedded solid core and a quiet central settling interval. Planet-sized clouds are not interchangeable with stellar seeds. The 96,000-km eligibility ceiling and the larger retained-atmosphere ceiling serve different purposes.

HUD READING / A grouped gas percentage is not the same as retained gas satisfying the first gate.

SOURCES S01,S05,S11 / register p.52

Compression without vanishing

Mass merges first; the extended envelope contracts toward its supported radius.

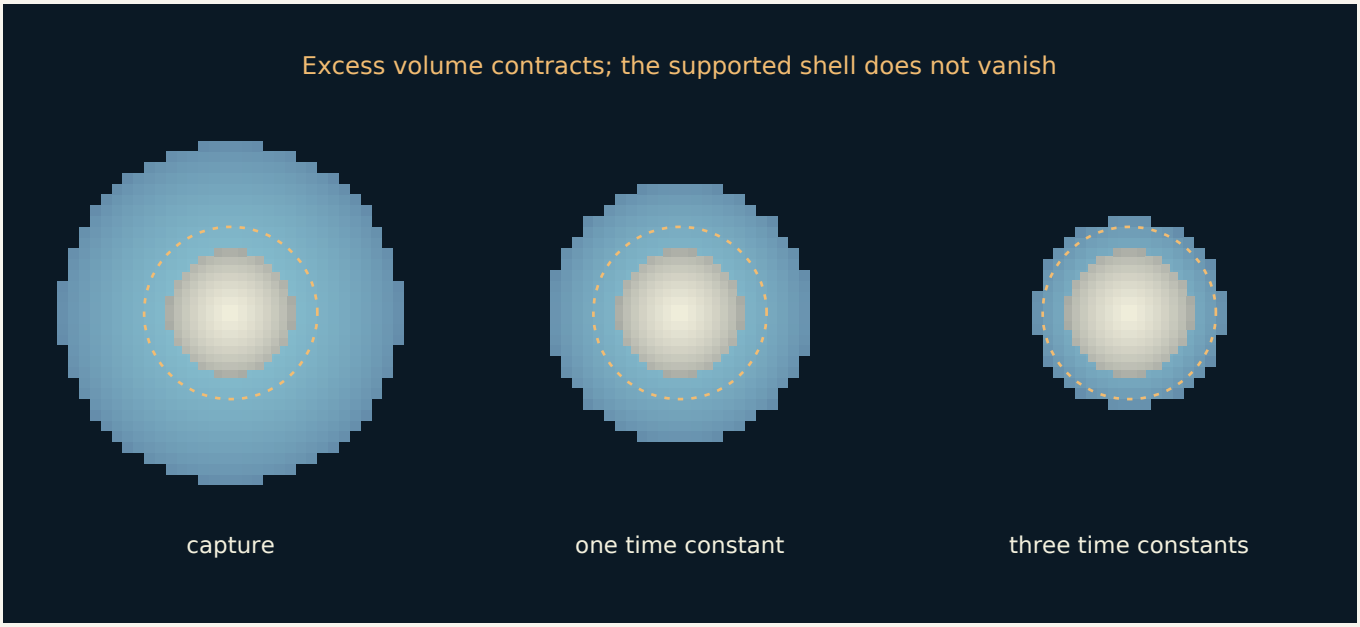


FIG. 24 / Illustrative snapshots of excess-volume decay; the solid core and shell floor remain.

$$R_{\mathrm{next}}^3 = R_{\mathrm{eq}}^3 + (R_{\mathrm{old}}^3 - R_{\mathrm{eq}}^3)e^{-\Delta t/\tau} \tag{24.1}$$

$$R_{\mathrm{eq}} \geq R_{\mathrm{core}} + 750\ \mathrm{km} \tag{24.2}$$

Start broad. Captured gas initially preserves an extended envelope. Its radius then relaxes toward an equilibrium radius derived from retained composition, compression factors and the supported planetary scale. It must not jump directly onto the visible solid.

Current volume rule. Hydrio, Helio, Carb and Oxio use a compression factor of four: the prescribed captured volume is one-quarter of their reference volume. This does not mean the diameter is divided by four.

Never erase the shell. The gas-bearing radius includes a minimum 750-km shell thickness around the core for those gases. The equilibrium and energy budget set additional limits; no step may contract below the supported target.

Time response. The envelope reference relaxation time is 18,000 universe seconds, or 15 real seconds at normal speed. It is an e-folding time for excess volume, not a promise that contraction finishes exactly at that instant.

ENERGY LIMIT / Spin and thermal budgets can stop a nominal contraction step before the formula's target.

SOURCES S01,S10,S11 / register p.52

A worked giant assembly

48,000-km Helio plus 6,000-km Irio, recalculated for v16 density.

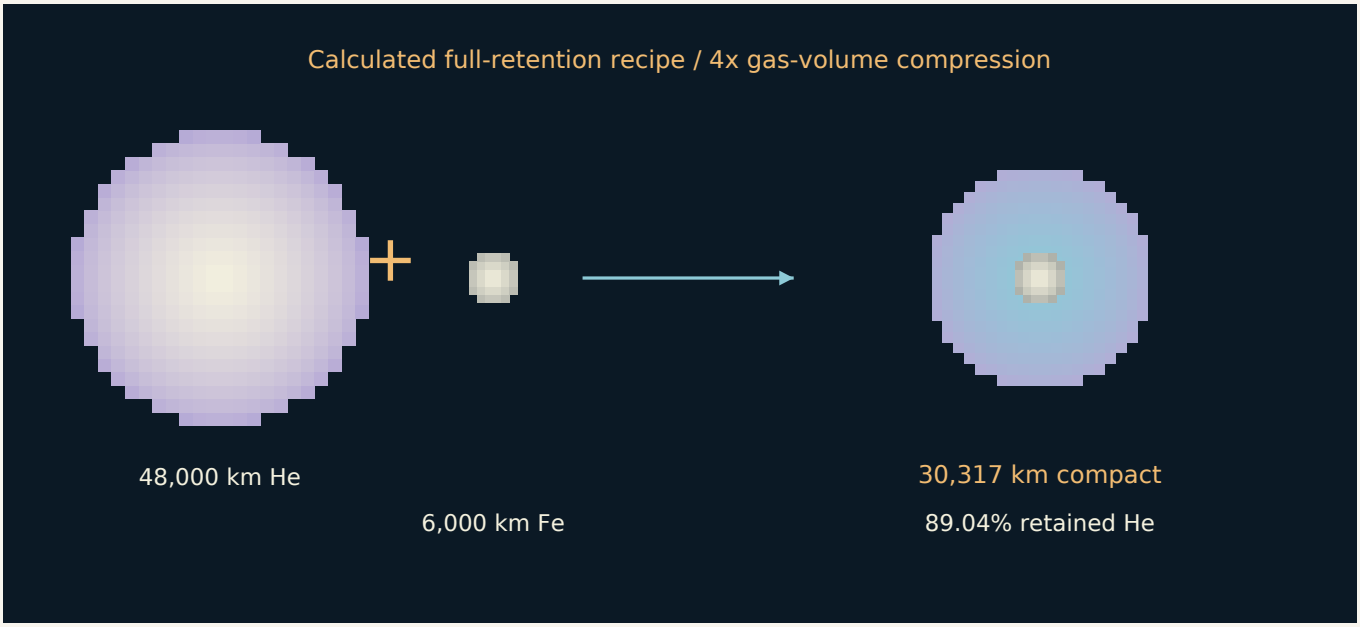


FIG. 25 / Calculated full-retention recipe; this is not a guarantee of capture in a live encounter.

$$D_{\text{compact}} = \left(6000^3 + \frac{48000^3}{4}\right)^{1/3} \text{km} \simeq 30{,}317 \text{ km} \tag{25.1}$$

$$f_{\text{He}} = \frac{125(48000)^3}{125(48000)^3 + 7874(6000)^3} \simeq 0.8904 \tag{25.2}$$

Inputs. The reference-density masses are about 7.24×10^{24} kg of Helio and 8.91×10^{23} kg of Irio. If all of both bodies is retained, the combined gas fraction is approximately 89.0%, comfortably above the GAS+ composition gate.

Compact volume. Applying the current fourfold gas-volume compression gives a compact diameter of approximately 30,317 km. That is just above the 30,000-km GIANT+ gate.

Why older observations differ. Density normalization, earlier compression factors, incomplete capture and grouped-but-unmerged inventory can all change the interpretation. The old 48,000-km screenshot is not a calibration result for this v16 recipe.

Encounter requirements. The solid still has to settle and the cloud has to remain eligible. Stripping or a surviving external reserve reduces the retained total. Check the actual one-body inventory and compact metric instead of assuming the entire spawn amount merged.

CALCULATED / Full retention, pure He and Fe, stated reference densities and the current compression factor.

SOURCES S02,S06,S10,S11 / register p.52

Mass-radius equilibrium

Self-gravity supplies a size target, not automatic expansion.

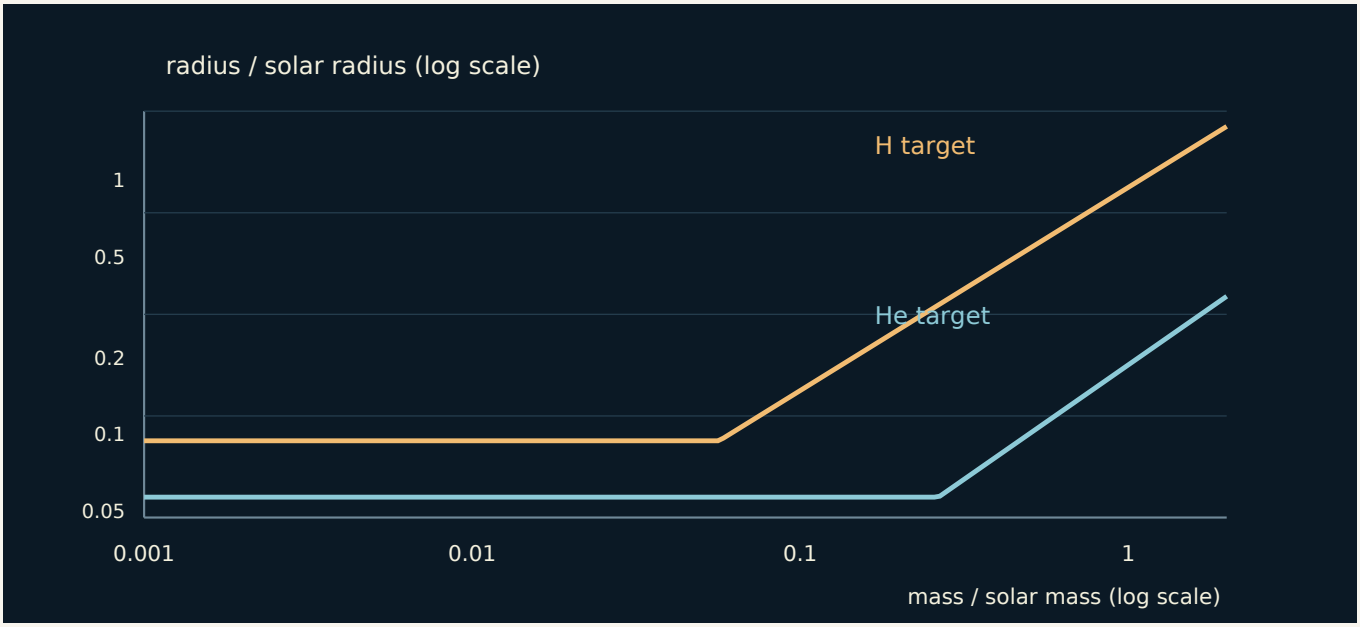


FIG. 26 / Calculated hydrogen and helium target-radius branches, shown logarithmically.

$$R_{\mathrm{H}} = \max\left(R_{\mathrm{J}}, R_{\odot}\,(M/M_{\odot})^{0.8}\right) \tag{26.1}$$

$$R_{\mathrm{He}} = \max\left(0.6R_{\mathrm{J}}, 0.2R_{\odot}\,(M/M_{\odot})^{0.9}\right) \tag{26.2}$$

Two anchors. The self-gravity model combines a planetary-radius plateau with a rising stellar mass-radius branch. Hydrogen and helium use different anchors. Composition interpolates logarithmically between the relevant limits.

Hydrogen branch. Its stellar radius follows the solar radius times mass in solar units to the power 0.8. Its low-mass plateau is the Jupiter reference radius, 69,911 km.

Helium branch. Its stellar target is 0.2 solar radii times the mass ratio to the power 0.9, with a plateau at 0.6 Jupiter radii. The larger of plateau and stellar target supplies the branch radius.

One-way relaxation. A cloud contracts only when it is larger than its supported target. A smaller cloud is not inflated to Jupiter's size just because the plateau exists. This is a gameplay-scale mass-radius proxy, not a full hydrostatic stellar structure solution.

CONSTANTS / R_J = 69,911 km; R_sun = 695,700 km; M_sun approximately 1.989×10^{30} kg.

SOURCES S02,S07 / register p.52

Contraction pays for heat

A smaller self-gravitating body releases binding energy.

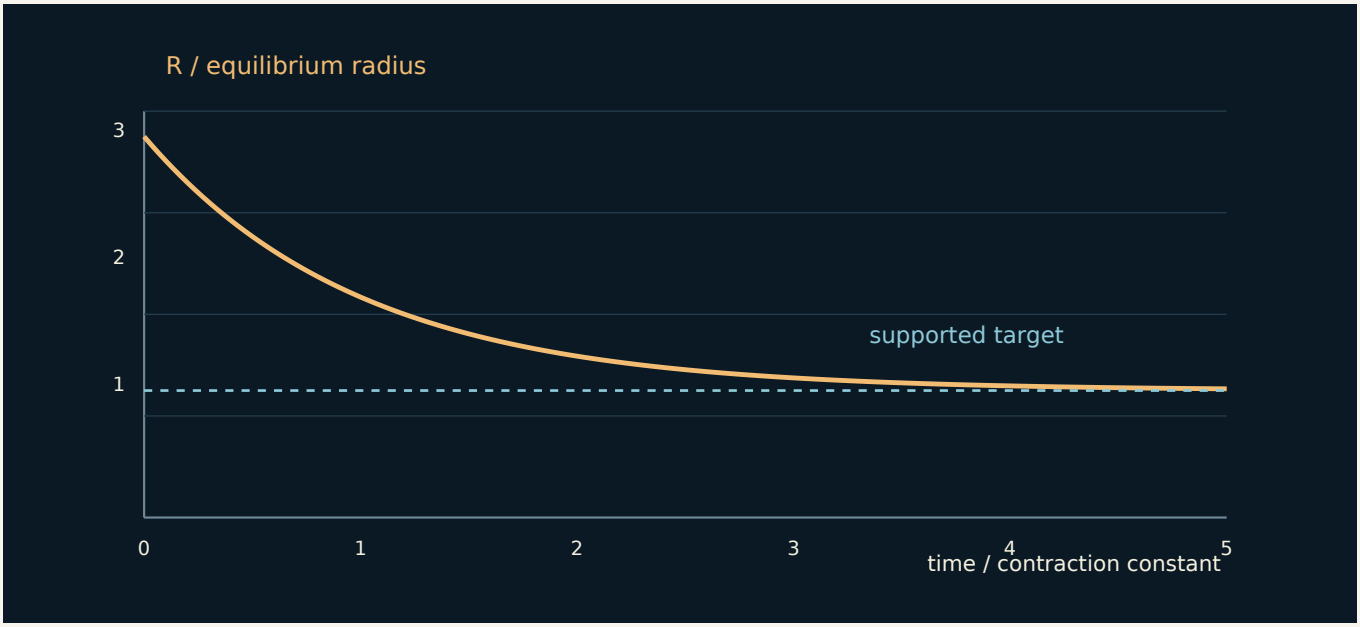


FIG. 27 / Calculated normalized contraction; released binding energy enters thermal accounts.

$$t_{\mathrm{ff}} = \sqrt{\frac{3\pi}{32G\bar{\rho}}}, \qquad \tau_{\mathrm{grav}} = 5t_{\mathrm{ff}} \tag{27.1}$$

$$U_{\mathrm{bind}} = -\frac{0.6GM^2}{\sqrt{R^2+\varepsilon^2}}, \qquad Q_{\mathrm{released}} = U_{\mathrm{old}} - U_{\mathrm{new}} \tag{27.2}$$

Relaxation clock. Self-gravitating contraction uses five free-fall times as its reference time scale. The free-fall estimate depends on mean density, so larger density changes the response without inserting a fixed universal animation timer.

Energy source. The softened binding estimate becomes more negative as a supported body contracts. The released difference can heat the matter. It is not an independent energy generator.

Partition. The current self-gravity model directs 15% of released binding energy toward the core, whose reference mass allocation is 20%. The rest supports the surrounding thermal state, subject to the actual reservoir bookkeeping.

Branch separation. This free-fall-controlled radius relaxation differs from the planetary excess-envelope volume relaxation described on page 24. They solve related but different tasks; their time constants should not be substituted for each other.

NO FREE HEAT / Any competing increase in spin or mechanical energy must also be paid.

SOURCES S02,S07 / register p.52

The Hydrio stellar branch

A pure-gas stellar transformation is not simply a hot collision.

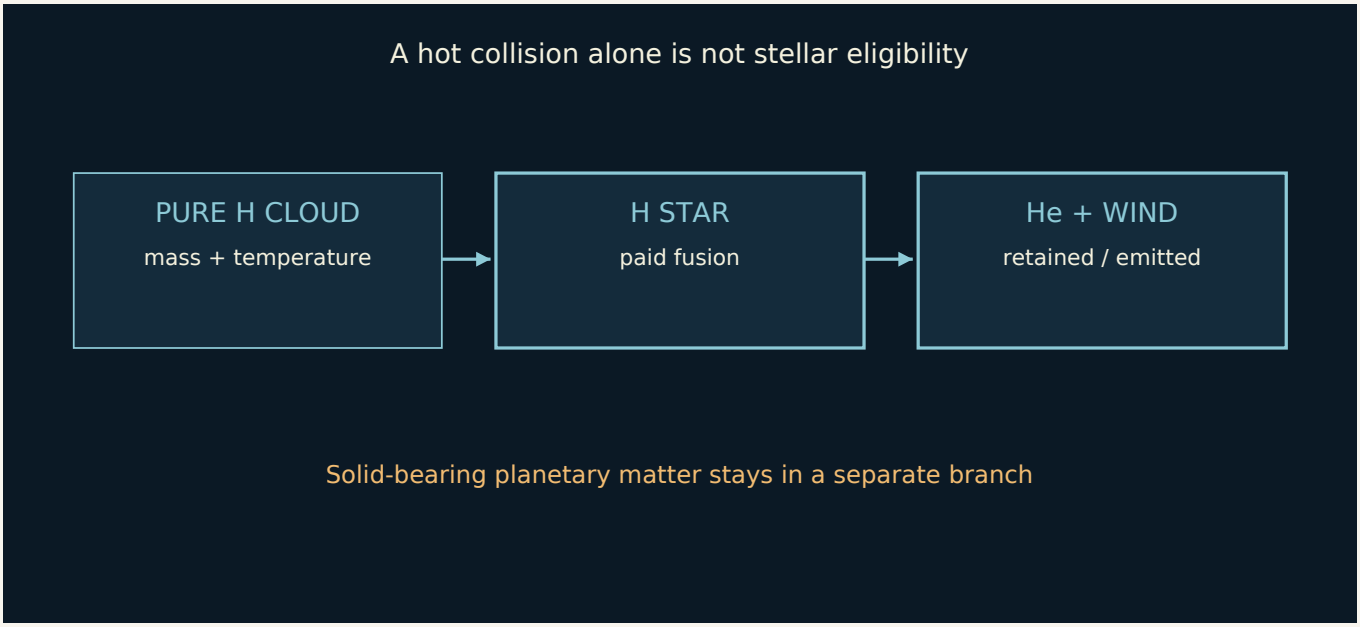


FIG. 28 / Hydrio cloud, contraction, eligibility and a paid Helio-producing stellar branch.

$$\mathrm{H} \longrightarrow \mathrm{He} + \text{energy} \tag{28.1}$$

$$Q_{\text{fusion}} = \delta\,\Delta m\,c^2, \qquad \delta_{\mathrm{H}} = 0.007 \tag{28.2}$$

Separate temperature from status. Every collision can heat matter. Stellar status additionally requires the gas-only branch, sufficient mass and the relevant core-temperature gate. A hot solid with gas around it remains a planetary composite.

Current gate. The Hydrio branch has a reference stellar mass threshold of 1.2×10^{29} kg and a core-temperature threshold of three million kelvin. Its fusion operating target is five million kelvin.

Material origin. Helio in this branch originates from consumed Hydrio rather than being spawned as an unrelated cosmetic element. Fusion energy is tied to the converted mass and a modeled 0.007 mass defect.

Inventory after emission. A star can emit retained fuel and reaction products through its separate wind process. The remaining HUD inventory must follow the retained matter, while any radiated energy or ejected packet is accounted for outside that inventory.

MODEL LABEL / H and He denote Kujira material inventories; this arrow abbreviates the reaction network.

SOURCES S01,S07,S12 / register p.52

Helio, Carb & Oxio

Products require the correct hot branch, density and reaction time.

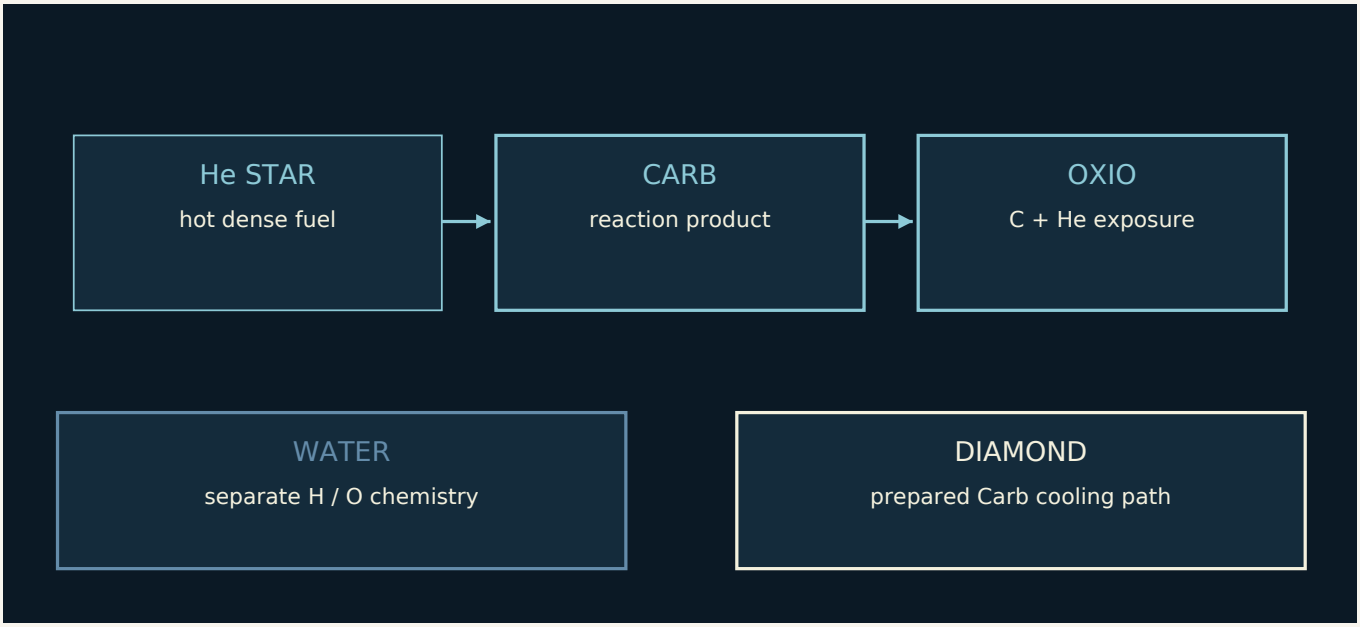


FIG. 29 / Stellar product branching contrasted with the separate water and diamond paths.

$$\mathrm{He} \longrightarrow \mathrm{C}, \qquad \mathrm{C} + \mathrm{He} \longrightarrow \mathrm{O} \tag{29.1}$$

$$Q_{\mathrm{He}} = 0.00065\,\Delta m\,c^2 \tag{29.2}$$

Helio star. The Helio branch uses a 6 × 10^{29}-kg mass threshold and a core-temperature gate of one hundred million kelvin. Its operating fusion target is one hundred fifty million kelvin. Its modeled mass defect is 0.00065.

Product sequence. The branch produces Carb, and the Carb-Helio path can produce Oxio only under suitable temperature, density and residence conditions. Cool touching parcels do not automatically become Oxio.

Distinct combinations. Hydrio-Oxio water chemistry is not nuclear fusion. Diamond formation is a separate prepared-Carb history and cooling artifact, not a universal rule that all cold Carb is diamond.

Conservation through conversion. A reaction moves material between species ledgers. Nuclear energy can include a declared rest-mass defect; chemical and phase changes use their own energy terms. The diagram is a branch map rather than a complete nuclear reaction catalogue.

READING RULE / Arrows show eligible modeled transformations, never automatic conversion on any collision.

SOURCES S01,S12 / register p.52

Winds, return & escape

Stellar ejection and ordinary gas-contact wakes are different mechanisms.

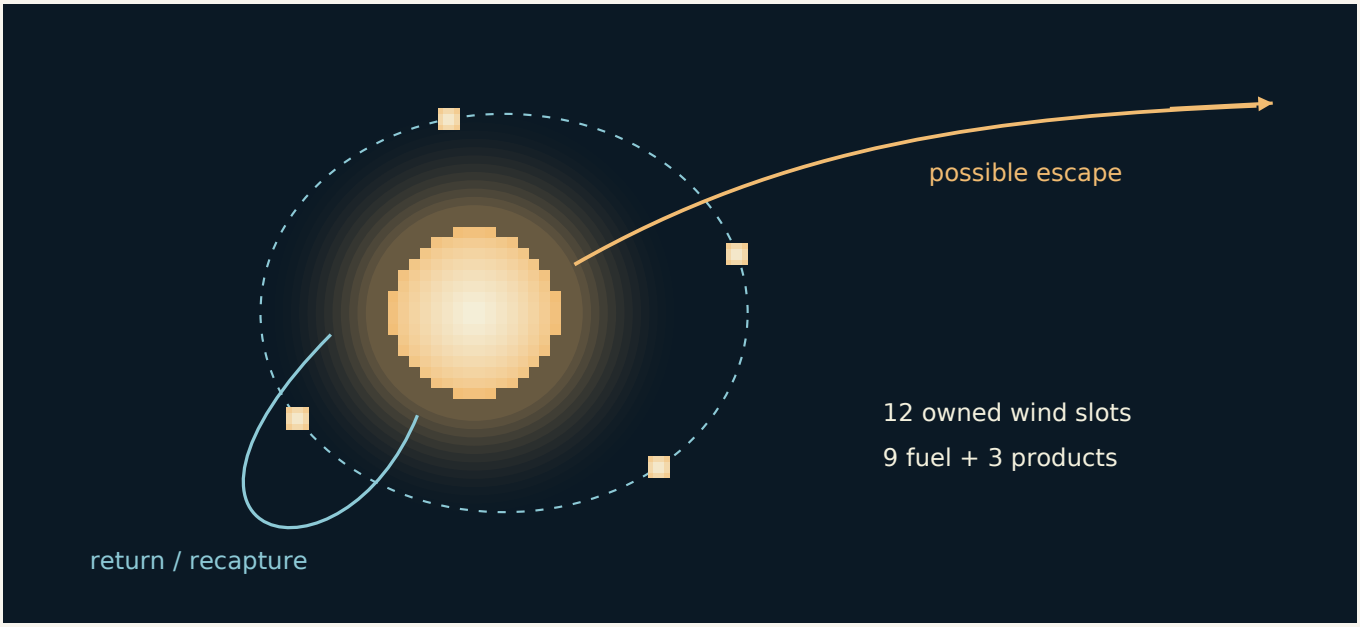


FIG. 30 / Paid stellar packets may return, orbit temporarily or leave the source's bound region.

$$v_{\mathrm{esc}}^2 = \frac{2G(M+m)}{\sqrt{r^2+\varepsilon^2}} \tag{30.1}$$

$$E_{\mathrm{specific}} = \tfrac{1}{2}|\mathbf{v}-\mathbf{v}_{\mathrm{host}}|^2 - \frac{G(M+m)}{\sqrt{r^2+\varepsilon^2}} \tag{30.2}$$

Bounded wind pool. The current design provides twelve wind slots per star: nine fuel and three product slots. Packets carry their source identity and elapsed wind time. The documented packet diameter range is 375 to 1,800 km.

Lifecycle. Winds begin with stellar eligibility. After 144,000 universe seconds, their special ownership expires and they become ordinary gas parcels. This lifetime is not a ban on subsequent physical interactions.

Return heating. A packet falling back can return kinetic and thermal energy. Its original launch and removed mass must already have been paid. Repeated emission and recapture must not create an unaccounted energy loop.

Escape frames. Escaping a planet is not the same as escaping its star. Compare velocity relative to the relevant host and use the same softened potential as gravity. Navigation speed belongs to the observer, not to the parcel's orbital ledger.

CAUTION / A negative two-body energy is a local diagnostic, not immunity to later multi-body encounters.

SOURCES S01,S03,S08,S12 / register p.52

Mapping a solar system

Light, heat, gravity and ownership are overlapping maps, not one radius.

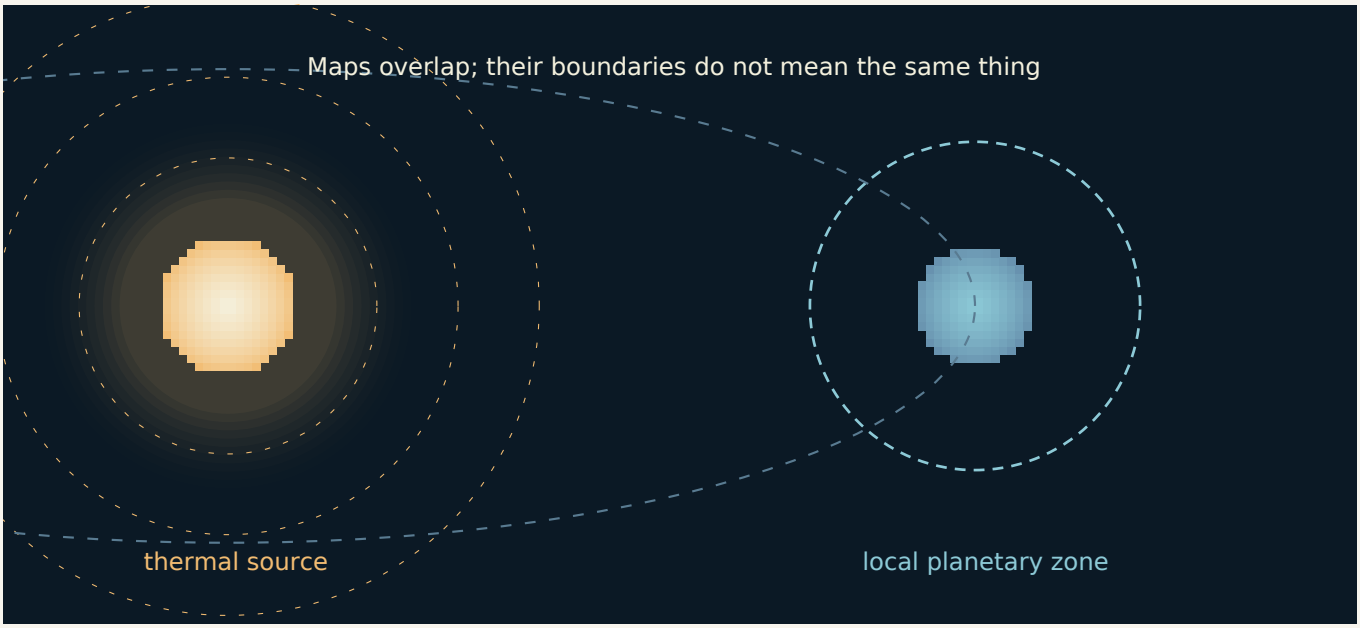


FIG. 31 / Schematic overlays: stellar heat field, planetary orbit, Hill zone and retained envelope.

Four different boundaries. The drawn luminous field explains the star's visible influence. The radiation model computes heating. Orbital zones describe local retention. A body's material envelope defines its actual extent. None should silently replace the others.

Planet membership. A system HUD counts qualifying solid or solid-gas bodies rather than every loose gas cloud. A nearby gas parcel remains material in motion, not a planet simply because it is visible inside the glow.

Dominant source. Orbital ownership uses the eligible heavier stellar source with the strongest gravitational pull. The hierarchy must be recomputed as objects move or their masses change.

Map reading. A large warm halo does not promise physical contact or immediate accretion. A planet well inside it may have a different retained atmosphere and cooling history from a newly spawned cloud at the same distance. These are different state variables, not inconsistent scales.

ILLUSTRATION / The circular overlays here are explanatory boundaries, not screenshots or measured orbital paths.

SOURCES S01,S03,S08,S14 / register p.52

The Hill neighbourhood

A planet's useful retention region is limited by the star.

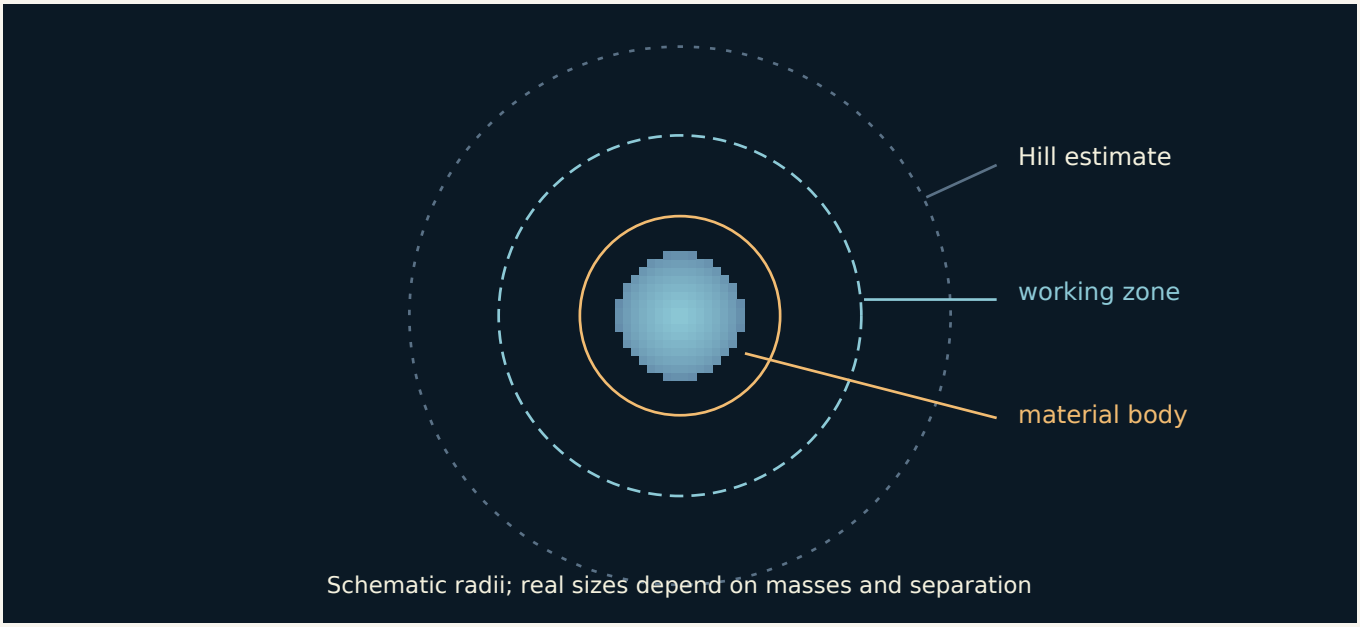


FIG. 32 / A schematic Hill radius, smaller working zone and the host's material surface.

$$r_{\mathrm{H}} = a\left(\frac{m_p}{3(M_\star + m_p)}\right)^{1/3} \tag{32.1}$$

$$r_{\mathrm{zone}} = \max(1.25R_p,\ 0.4r_{\mathrm{H}}) \tag{32.2}$$

Stellar competition. The Hill estimate relates orbital separation to the cube root of the planet-to-star mass ratio. It supplies a scale for where the planet's local gravity can compete with the stellar tide.

Working zone. The implementation uses 40% of the Hill radius with a minimum of 1.25 host radii. Without a stellar host, its fallback is sixty host radii. These are practical zone policies, not additional attractive forces.

Interpret carefully. The minimum-radius floor means a tiny or strongly stressed body can have a policy zone even when the Hill approximation becomes restrictive. Bound-energy and launch checks remain necessary.

Tide proxy. The documented tide-ratio expression reaches about 23.44 at 0.4 Hill radii. It is a diagnostic for the reduced model, not a complete tidal tensor or a general guarantee of orbit stability.

SCALING / Double the orbital separation and this Hill estimate doubles; double planet mass and it grows by a cube root.

SOURCES S03,S08 / register p.52

Nested orbital ownership

A moon belongs to a planet; that planet can belong to a star.

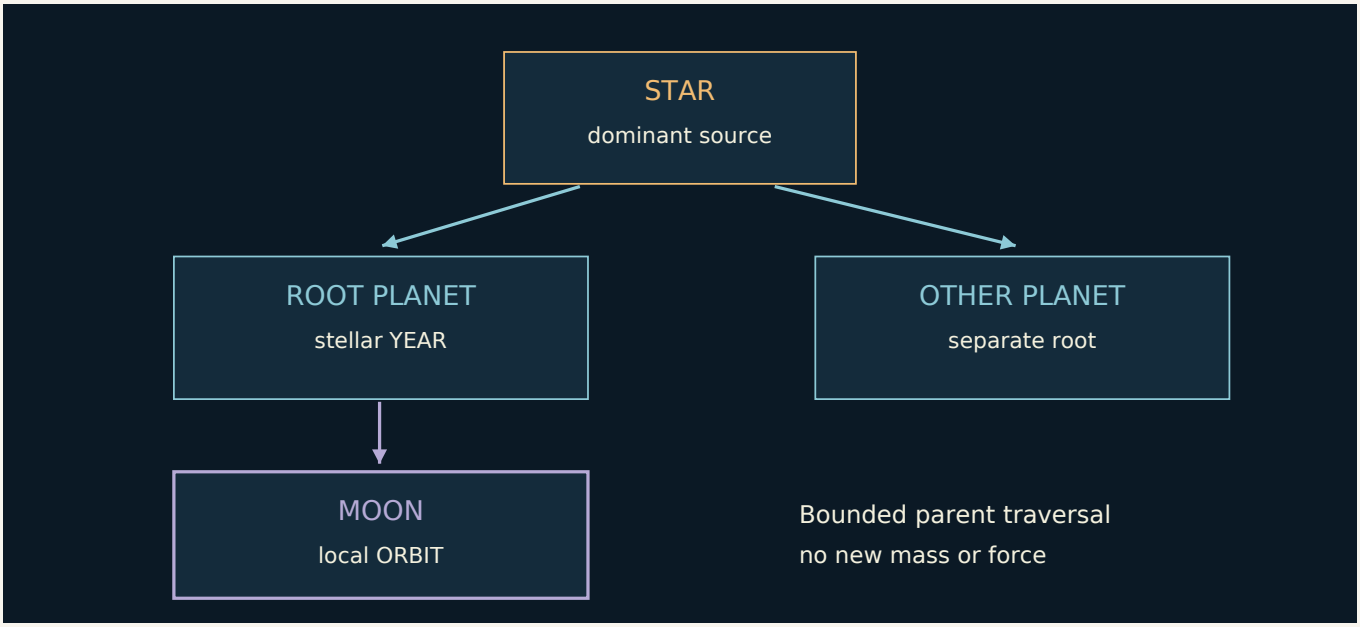


FIG. 33 / A bounded ownership tree with local motion carried through its parent hierarchy.

$$\mathbf{v}'_{\text{child}} = \mathbf{v}_{\text{child}} + \Delta\mathbf{v}_{\text{root}} \tag{33.1}$$

$$\Delta\mathbf{P}_{\text{star}} = -\Delta\mathbf{P}_{\text{owned group}} \tag{33.2}$$

Root and parent. A bound body selects an eligible more massive host. A moon's immediate host is a planet, while the root planetary system follows its dominant star. The documented traversal is bounded to four parent hops.

Carried guidance. When root-planet stellar guidance adjusts the root motion, owned bodies inherit the corresponding translation. Stellar recoil balances the internal momentum exchange rather than treating the star as a fixed external rail.

Nested Hill zones. A moon's local zone is constrained by its parent-planet context and the wider stellar Hill limit. A hierarchy therefore cannot manufacture unlimited retention space by stacking arbitrary parent labels.

HUD consequence. The moon's ORBIT describes its immediate parent-relative period. Its YEAR is associated with the root planet's stellar orbit. Those two clocks answer different questions and should not be collapsed into a single speed label.

BOUND / Ownership is derived state and must be reconsidered after escape, merger or host removal.

SOURCES S03,S04,S08,S09 / register p.52

A bounded launch window

Debris can leave a surface without being launched out of its local system.

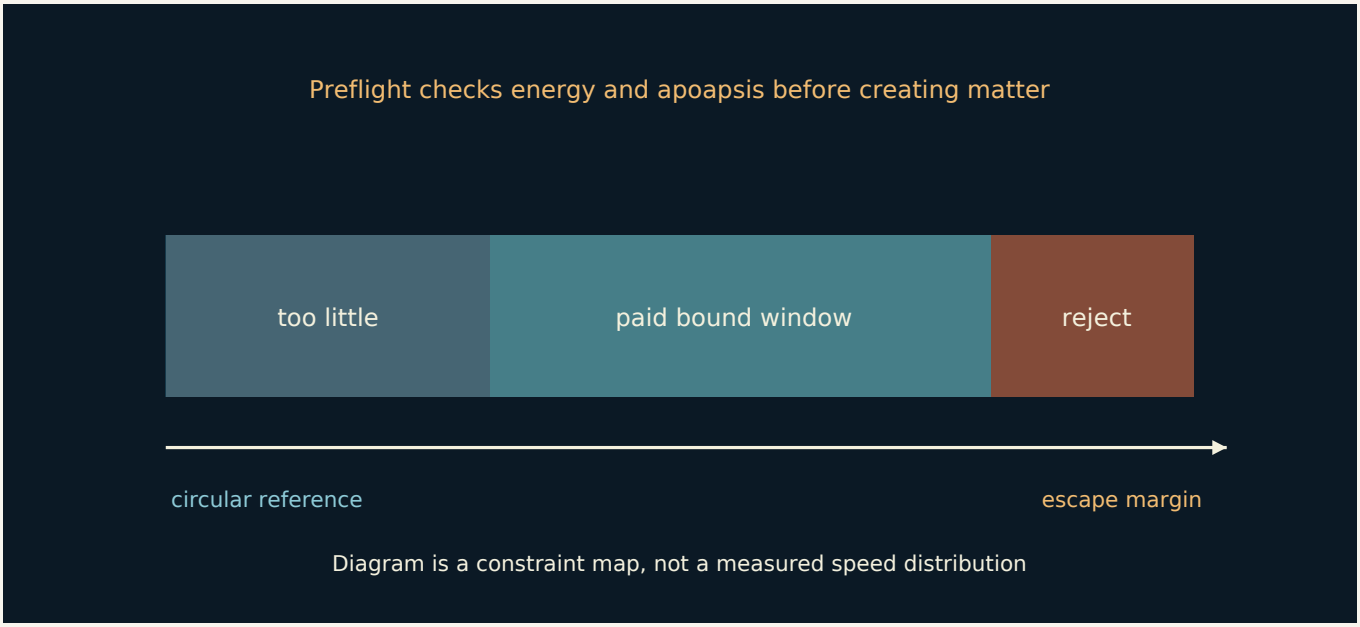


FIG. 34 / Illustrative speed interval from circular motion to a zone-limited escape margin.

$$v_c = \sqrt{\frac{GM_p}{r_L}}, \qquad v_{\max} \leq 0.98 v_{\mathrm{esc}} \tag{34.1}$$

$$v_{\max} \text{ also constrained by } v_{\mathrm{esc}} \sqrt{\frac{r_{\mathrm{zone}}}{r_L + r_{\mathrm{zone}}}} \tag{34.2}$$

Eligible host. Belt-capable bodies can launch suitable small debris into a bound local region. A paid tangential launch makes the insertion point approximately periapsis. The selection uses available energy and a seeded distribution within an allowed speed range.

Upper edge. The launch ceiling combines a margin below local escape with a radius-based zone constraint. Circular speed is the lower reference, not a command to add energy that the collision does not possess.

Preflight. The complete proposed event is checked before committing it. A fragment with positive host-relative energy or an apoapsis outside the zone invalidates the bound insertion. Rejected material stays in the ledger instead of vanishing.

Low energy. When the event cannot afford circular insertion, the model takes the appropriate lower-energy radial response. These restrictions apply to this specific debris branch; they do not rewrite all free-body orbital dynamics.

SCOPE / The complete helper includes a circular-speed floor and preflight; this displays its main constraints.

SOURCES S03,S08 / register p.52

Imminent accretion

A capture buffer should guide an eligible inward path, not teleport every flyby.

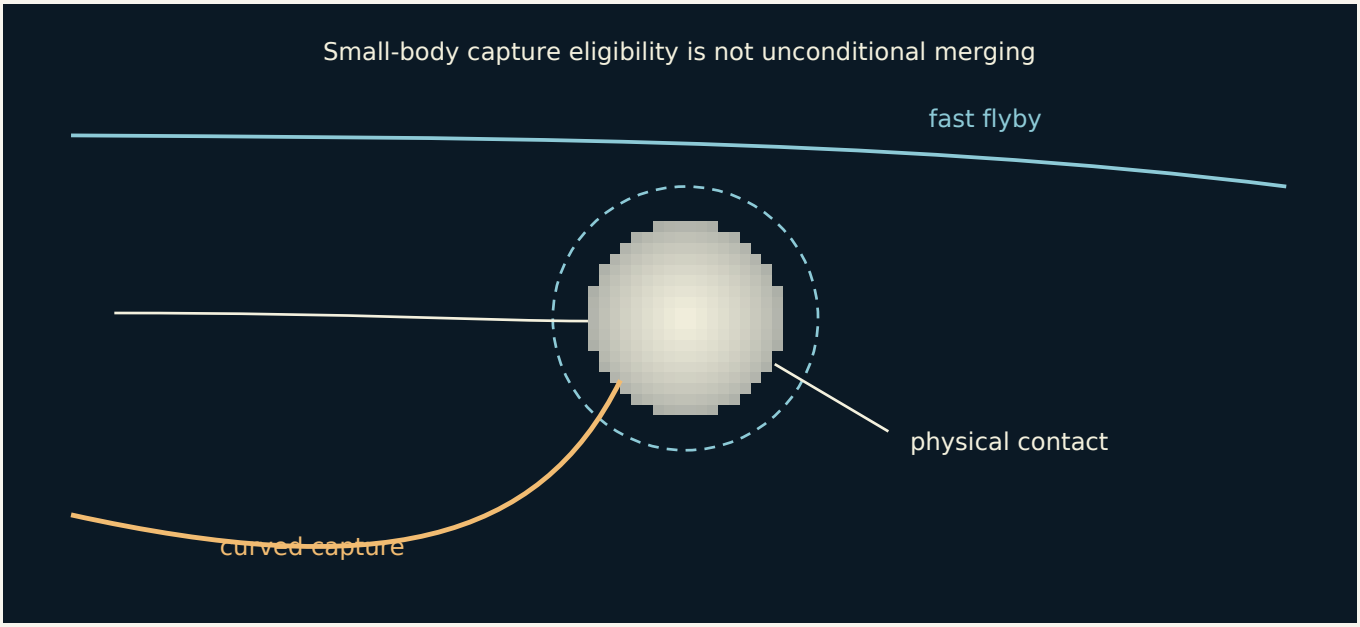


FIG. 35 / Distinct schematic outcomes: free flyby, buffer-crossing capture and physical contact.

$$R_{\text{buffer}} = 1.35R_p, \qquad m_{\text{small}} \le 0.10M_p \tag{35.1}$$

$$r_{\text{peri}} < R_{\text{buffer}}, \qquad r_{\text{apo}} \le r_{\text{zone}} \tag{35.2}$$

Buffer geometry. The planetary reference buffer extends to 1.35 host radii. Special small-body handling applies to bodies no more than 10% of host mass. Comparable planets still use ordinary gravity and contact rules.

Curved capture. For an eligible bound orbit whose periapsis crosses the buffer and whose apoapsis fits the zone, guidance promotes a contacting trajectory. The intent is to avoid persistent near-surface circulation that never resolves.

No arbitrary countdown. Contact follows the geometric and energetic situation, not a global ten-second destruction timer. A genuinely fast unbound flyby is not forced into a merge simply because its path crosses a drawn neighbourhood.

Insertion grace. Newly inserted bound debris receives a documented six launch-orbit-period grace before buffer capture can reclaim it. Actual material contact remains active throughout; grace does not let a chip pass through a solid core.

NOT A FORCE LAW / These are eligibility conditions around the gravitational solver, not Newtonian gravity itself.

SOURCES S01,S03,S08 / register p.52

Dissipation into a disk

Remove radial motion without supplying a new tangential kick.

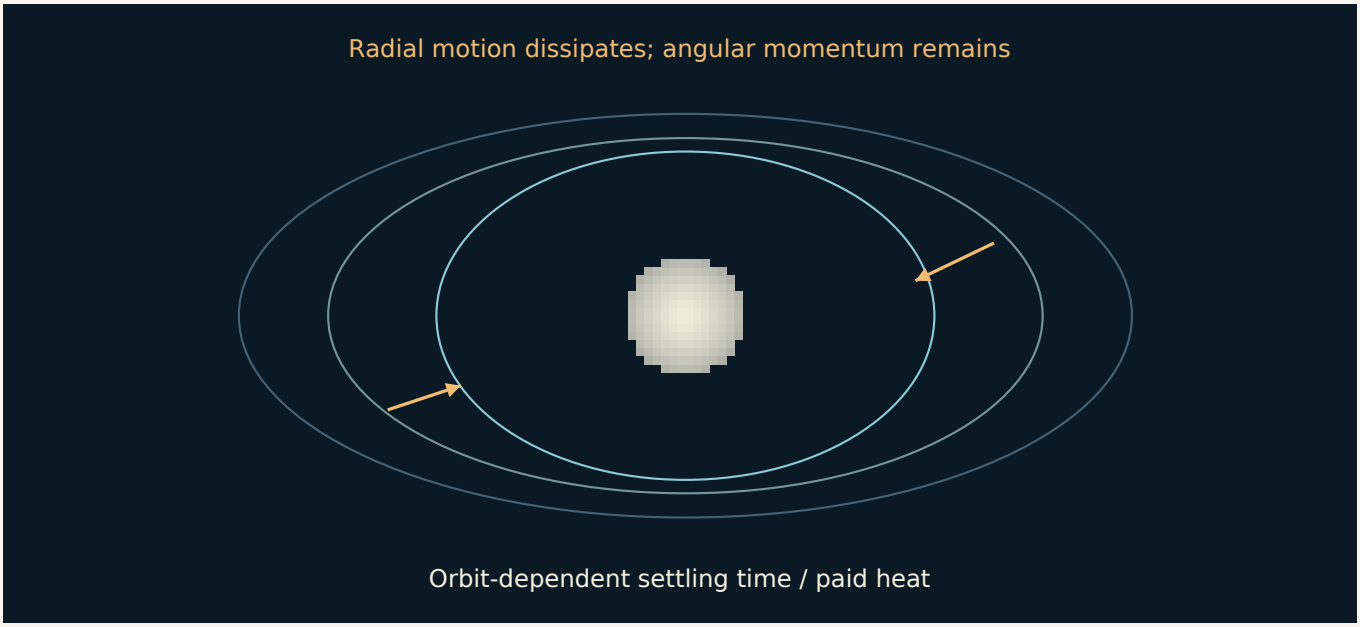


FIG. 36 / Illustrative radial-velocity decay while angular momentum remains in the orbit.

$$r_c = \frac{h^2}{\mu_G}, \qquad \mu_G = G(M+m), \qquad \tau_{\text{set}} = 2P_K(r_c) \tag{36.1}$$

$$v_r' = v_r e^{-\Delta t/\tau_{\text{set}}}, \qquad Q = \tfrac{1}{2}\mu_{\text{red}}(v_r^2 - v_r'^2) \tag{36.2}$$

Targeted damping. The settling model damps the relative radial velocity. Equal and opposite radial impulses preserve pair angular momentum while removing radial kinetic energy. That removed energy is converted to heat.

Orbit-dependent clock. The damping time is two Kepler periods at the circularization radius derived from specific angular momentum. This ties the response to the local orbit rather than to display zoom or frame count.

Eligibility. Small participants, at most 10% of host mass, enter this branch only in the appropriate region. Solids settle outside the material envelope; gas may settle within capture reach. Belt-host restrictions still matter.

Do not rescue a dive. A body already on a valid inward capture path must not be circularized back out of the accretion buffer. Disk formation is therefore coordinated with contact prediction, not applied indiscriminately to every nearby object.

NOTATION / mu_G is a gravitational parameter; mu_red is reduced mass. Their units and roles differ.

SOURCES S03,S08 / register p.52

Roche bands & moon space

Tidal disruption and accretion do not share the same inner edge.

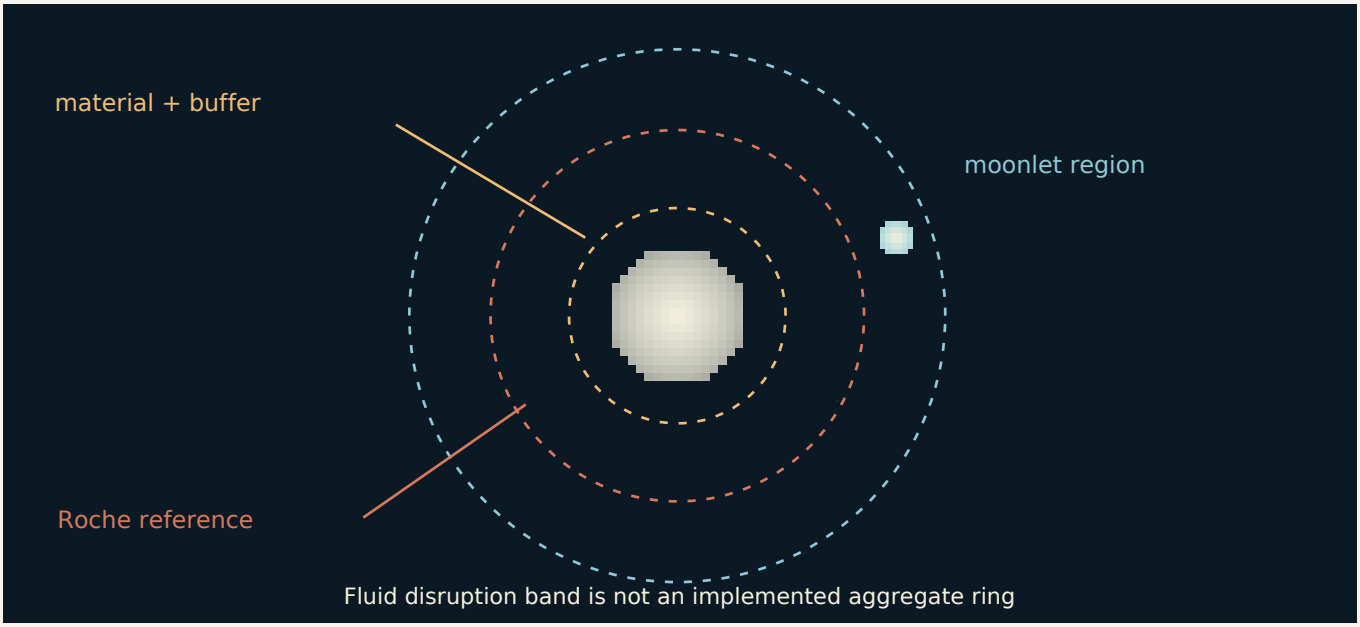


FIG. 37 / Schematic surface, fluid Roche band and admissible outer moonlet region.

$$r_{\text{Roche, fluid}} = 2.44 R_p \left(\frac{\rho_p}{\rho_d}\right)^{1/3} \tag{37.1}$$

$$r_{\text{Roche, rigid}} = 1.26 R_p \left(\frac{\rho_p}{\rho_d}\right)^{1/3} \tag{37.2}$$

Density-sensitive scale. The model evaluates fluid and rigid Roche-radius estimates from host radius and the cube root of host-to-debris density. Different fragment materials therefore have different disruption reference radii.

Moon inner edge. A candidate moonlet must lie beyond the relevant Roche boundary, capture buffer and material envelope, with its own radius included. Clearing only the visible core is insufficient.

Inside the band. The inner disrupted region motivates the ring roadmap. It should not be described as an implemented continuous ring fluid: the current system still consists of discrete bodies and parcels.

Approximation. The coefficients are conventional reduced-body estimates applied in a pixelated model. Strength, deformation and multi-body tides are not resolved continuously. The diagram labels a useful decision map, not an exact fracture simulation.

MAP RULE / A safe-looking orbital radius can still fail envelope, energy or ownership checks.

SOURCES S03,S08 / register p.52

Roche overflow

An overextended atmosphere can transfer through a tidal bottleneck.

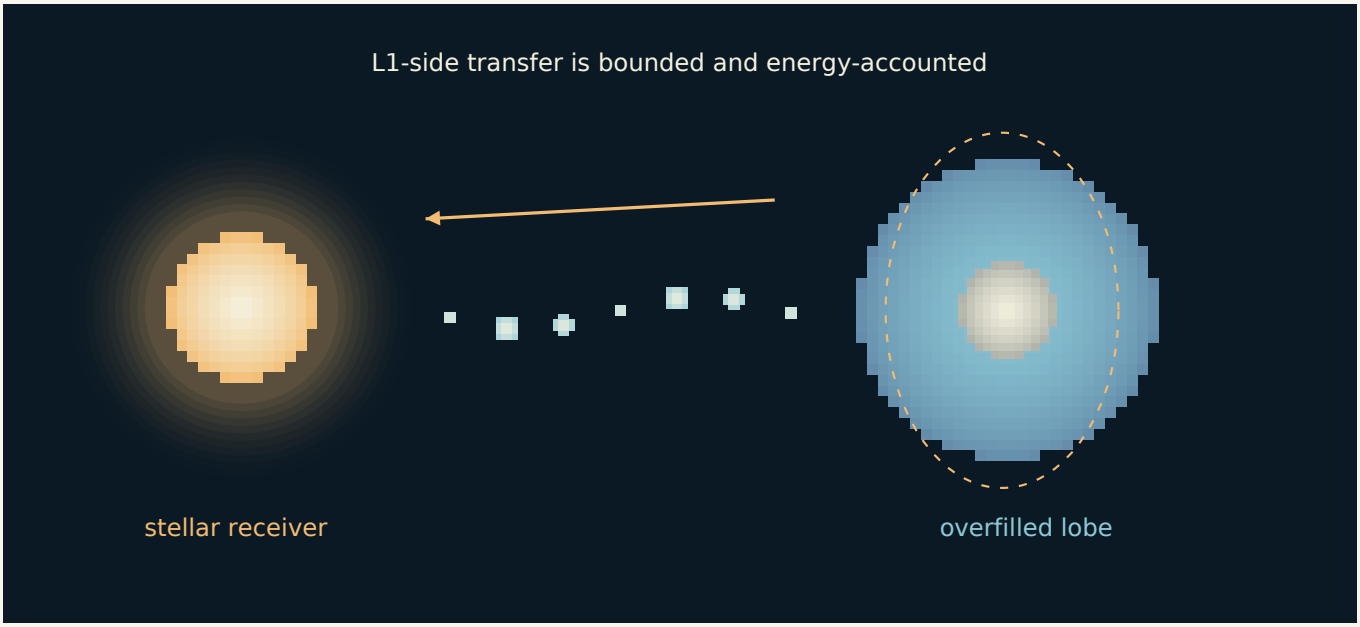


FIG. 38 / Schematic Roche lobe and L1-side parcel transfer toward the stellar host.

$$\frac{r_L}{a} = \frac{0.49q^{2/3}}{0.6q^{2/3} + \ln(1+q^{1/3})}, \qquad q = \frac{m}{M} \tag{38.1}$$

$$t_{\mathrm{dyn}} = \sqrt{\frac{R^3}{Gm}} \tag{38.2}$$

Lobe estimate. The Eggleton approximation supplies the body's effective Roche-lobe radius relative to separation. When the envelope exceeds that scale, excess gas can be stripped in bounded parcel events.

Cadence and amount. The current model allows an L1-side parcel at half a dynamical time. It starts from half the outside-shell volume, limits the event share to 0.2%-8%, and caps the parcel radius at 40% of the envelope radius.

Payment. Lift, launch work and any available external tidal release enter the budget. Crossing a geometric boundary alone does not authorize an arbitrary amount of new outward kinetic energy.

Destination. Gas inside the stellar stripping region can transfer toward the star; farther out it may form a multi-parcel torus. The handoff reports roughly forty parcels in a demanding overflow case, an open crowding concern rather than a finished aggregate-ring optimization.

PERFORMANCE / Explicit parcel overflow is implemented; an efficient aggregate annulus is still proposed.

SOURCES S01,S03,S08 / register p.52

Who may build a belt?

The belt threshold is tied to a compact giant reference mass.

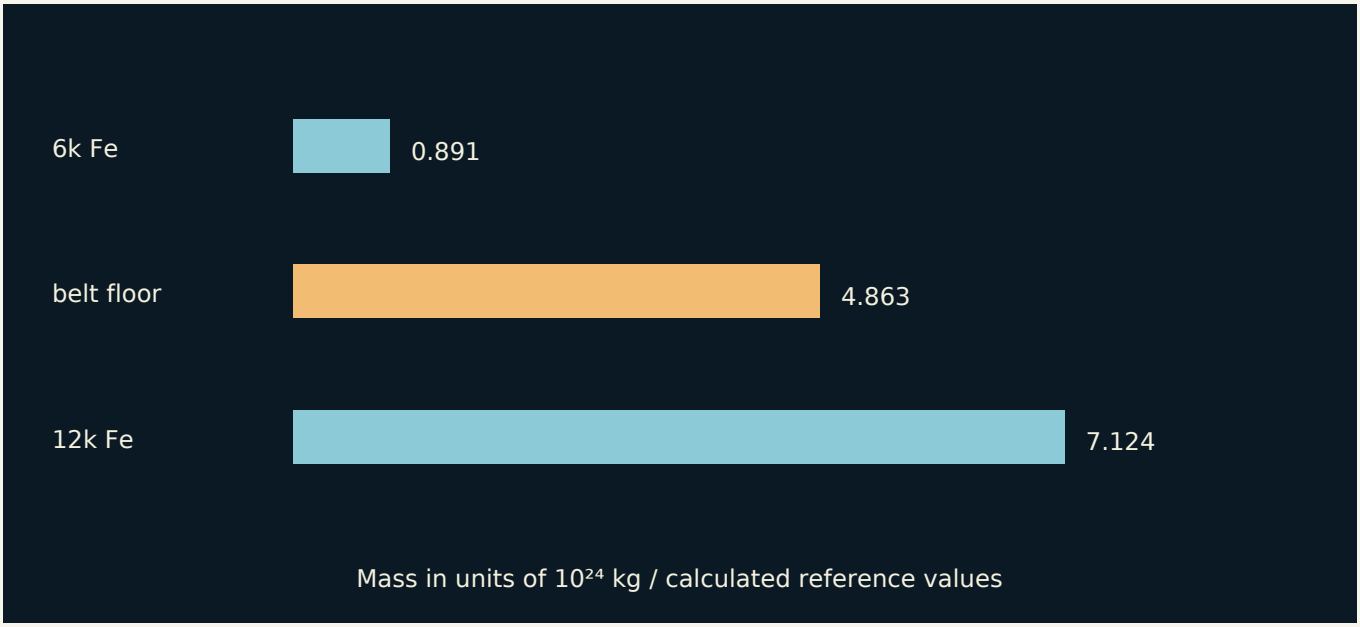


FIG. 39 / Calculated host masses compared with the lightest compact-giant belt threshold.

$$M_{\text{belt}} = 4\rho_{\text{H}}\tfrac{4\pi}{3}(15{,}000\ \text{km})^3 \simeq 4.86\times10^{24}\ \text{kg} \tag{39.1}$$

Reference host. The current belt threshold is the mass of a 30,000-km compact Hydrio giant using fourfold reference gas density. This gives approximately 4.86×10^{24} kg.

Consequences. A 6,000-km Irio body has about 8.91×10^{23} kg and lies below the threshold. A 12,000-km Irio body has about 7.12×10^{24} kg and lies above it. Diameter alone does not determine belt eligibility.

Below threshold. Small-host debris receives a compact return-oriented radial bound, with a maximum apoapsis constrained by 90% of the zone and four host radii. That avoids giving every tiny rock a durable satellite swarm.

Above threshold. A qualifying host can participate in belt insertion, disk settling and moonlet assembly, still subject to the local Roche, contact, energy and zone checks. Threshold crossing is necessary, not sufficient, for a stable moon system.

CALCULATED / Convert the radius to metres before applying the 86 kg/m^3 Hydrio reference density.

SOURCES S02,S03,S06,S08 / register p.52

From neighbours to moonlets

An unstable close pair can coalesce without losing its ledger.

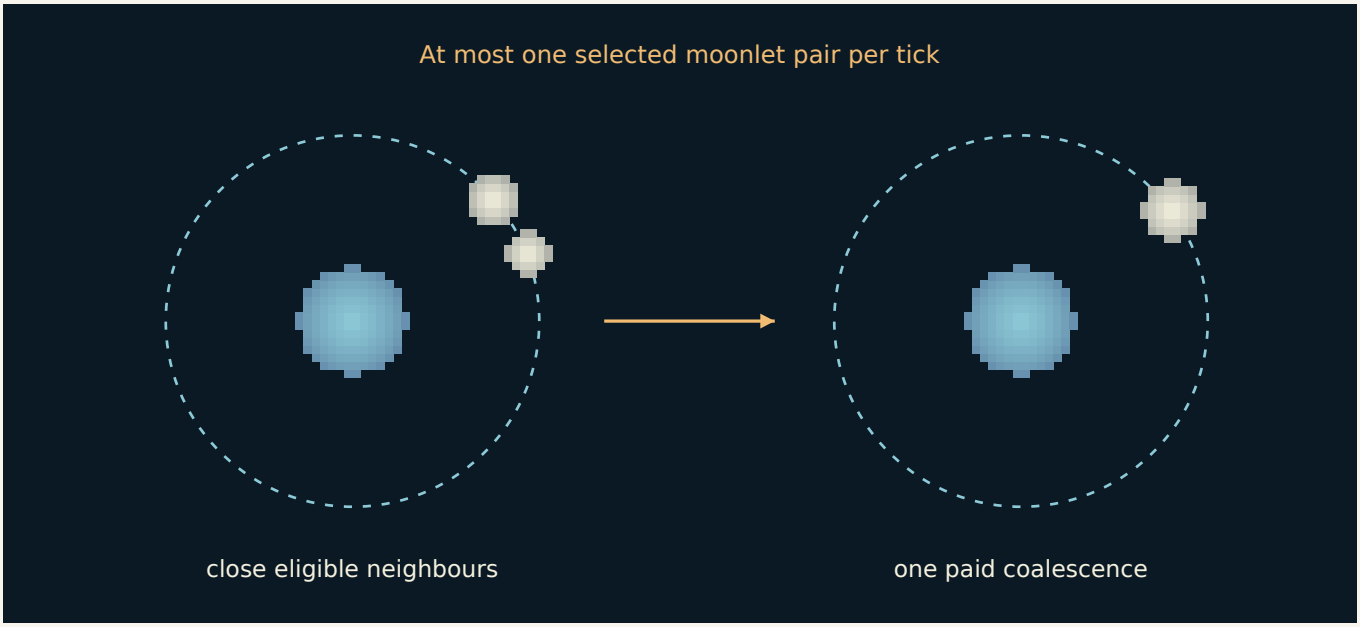


FIG. 40 / Illustrative pair spacing against a mutual-Hill scale; not a universal stability proof.

$$R_{\mathrm{H,\,mut}} = \left(\frac{m_1 + m_2}{3M}\right)^{1/3} \frac{a_1 + a_2}{2} \tag{40.1}$$

$$\Delta a < 2\sqrt{3}\, R_{\mathrm{H,\,mut}} \tag{40.2}$$

Shared parent. Eligible moonlet candidates orbit the same host, are each at most 10% of its mass, and remain outside the prohibited inner bands. Their semimajor axes and masses define a mutual-Hill spacing estimate.

Bounded work. The assembly rule selects the most unstable eligible pair and processes at most one pair per tick. It does not repeatedly merge an entire crowded belt inside one unbounded loop.

Coalescence budget. The remnant's material, translational motion, spin and heat are reconciled. Sideways relative angular momentum can be transferred to the host's spin, with other potential-energy changes included in the transaction.

Interpretation. The 2 square-root-of-3 mutual-Hill criterion motivates this reduced spacing rule. It is not a theorem guaranteeing all eccentric or multi-moon configurations are stable. Live orbital acceptance remains a separate gameplay question.

STATUS / Pair assembly is implemented; a completed continuous ring and tidal migration model are not.

SOURCES S01,S03,S08 / register p.52

Rings: the next representation

A proposed aggregate annulus is not yet a current entity.

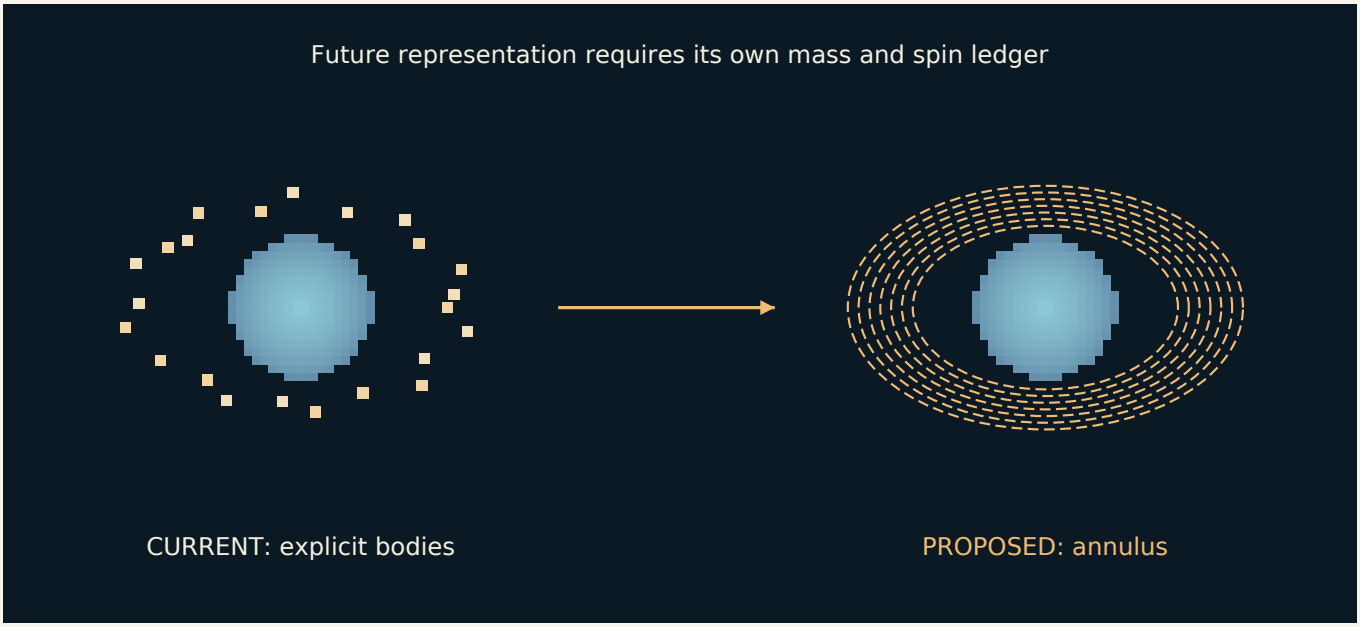


FIG. 41 / Proposed evolution from many parcels to an aggregate ring with explicit inventory.

$$M_{\mathrm{ring}} = \sum_s m_s, \qquad L_{\mathrm{ring}} = \int r\, v_\theta \,\mathrm{d}m \tag{41.1}$$

Current boundary. Discrete debris, tidal stripping, local settling and moonlet assembly provide ingredients for rings. The v16.3.1 snapshot does not yet contain the proposed single annulus entity that would replace many orbiting parcels.

Proposed representation. An aggregate ring would need retained species mass, inner and outer radii, angular momentum, thermal state and an explicit host. Its displayed opacity would follow its inventory rather than a free decorative texture.

Required interactions. Arrivals would exchange mass and momentum with the annulus; spreading would change its radii; losses and moonlet formation would debit its ledger. A ring cap would limit representation cost without deleting material.

Migration obligation. A new persistent ring representation would require a future save-schema change and migration coverage. Existing parcels must not be silently reclassified as a finished ring. The original visual ambition remains a small bounded set of clear, pixelated orbital layers.

PROPOSED / This page describes a development direction, not a claim that aggregate rings ship in v16.3.1.

SOURCES S01,S03 / register p.52

Universe time & warp

Warp schedules more fixed steps; it does not enlarge the physical time step.

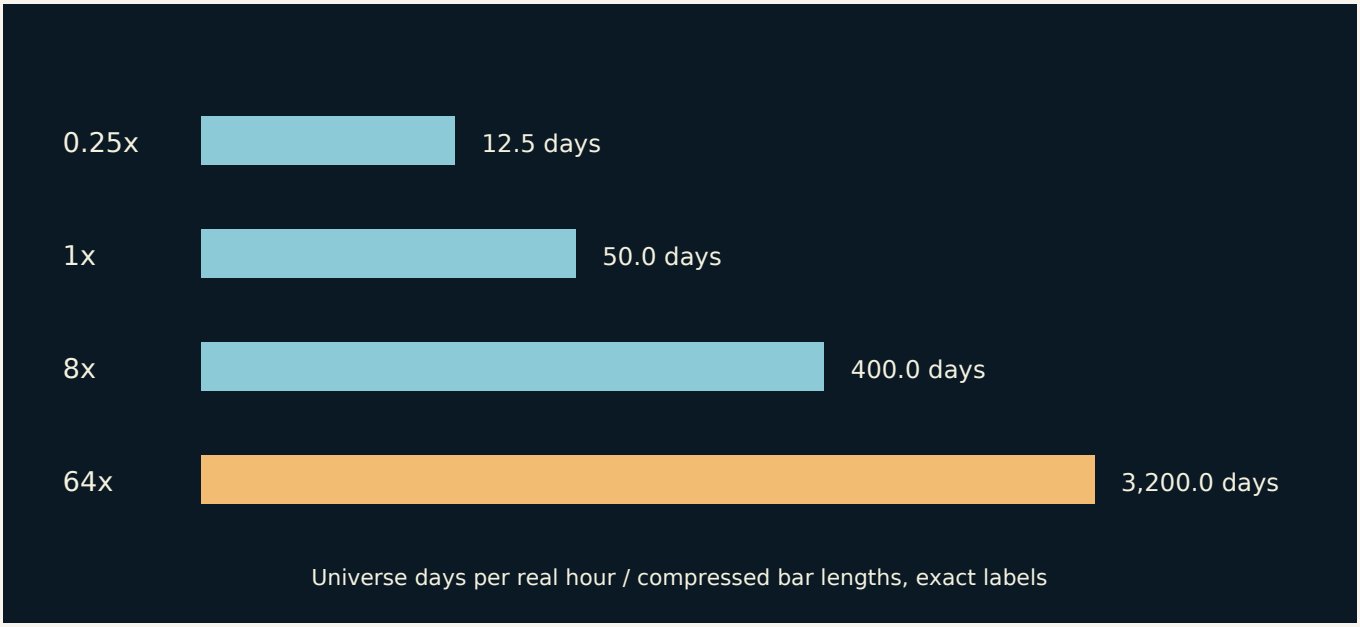


FIG. 42 / Calculated universe time advanced during one real hour, across the warp range.

$$t_U = N_{\text{ticks}}\,(10\ \text{s}) \tag{42.1}$$

$$\frac{\mathrm{d}t_U}{\mathrm{d}t_{\text{real}}} = 1200w, \qquad 0.25 \le w \le 64 \tag{42.2}$$

Fixed step. The simulation advances in ten-second universe ticks. At normal speed, one real second corresponds to 1,200 universe seconds: twenty simulated minutes. Warp ranges from one-quarter speed to sixty-four times normal.

Budgeted scheduling. More warp requests more ticks. The documented tick ceiling, frame clamp and eight-millisecond physics budget prevent a slow frame from turning into an unlimited catch-up loop.

Conversions. One real hour at normal speed represents fifty universe days. At 64x it represents 3,200 days, or approximately 8.76 Julian years, if the requested rate is sustained. This arithmetic corrects an older illustrative time example.

Freeze and pause. Universe freeze stops matter and its clock while preserving permitted observation and navigation. It differs from the ordinary menu pause. A higher camera speed must not be confused with a faster universe clock.

RATE QUALIFIER / Actual progress is constrained by frame scheduling and available compute time.

SOURCES S04,S09 / register p.52

Age belongs to a body

Birth history is persisted; unavailable history is not invented.

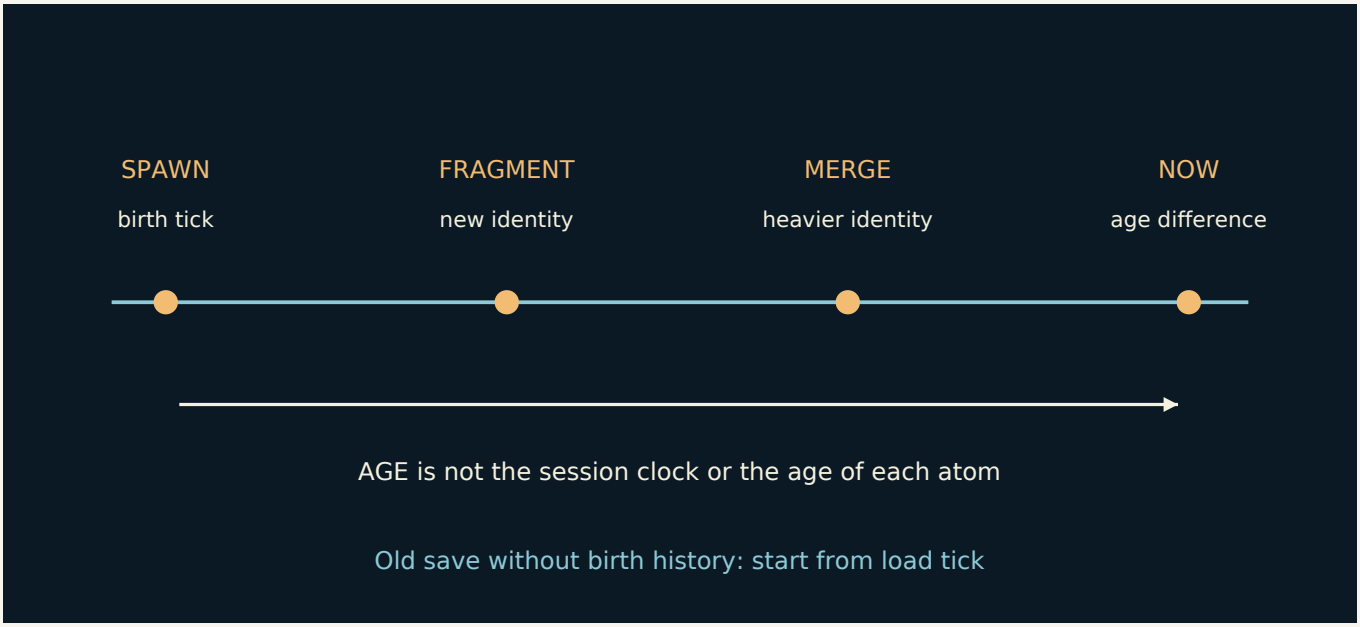


FIG. 43 / A body timeline through spawn, fragmentation, merger and an older-save migration.

$$t_{\mathrm{age}} = (N_{\mathrm{now}} - N_{\mathrm{born}})\,\Delta t_{\mathrm{tick}} \tag{43.1}$$

Birth tick. Each body stores when it entered the modeled universe. AGE is the difference between the current tick and that birth tick, multiplied by the fixed step. It is not how long the user has been watching the body.

New fragments. An emitted or impact-created parcel starts its own birth record. Its age does not automatically equal the age of the star or parent body from which its material originated.

Continuity choices. A merger retains the heavier participant's birth reference under the current identity policy. A gas cloud becoming a star keeps its cloud age. These choices document object identity, not the age of individual atoms.

Migration. Earlier v15 saves have no reliable stored birth history. On migration their bodies start from the load tick. Displaying a young migrated body is more honest than reconstructing a fictional formation date.

INTERPRETATION / Body age, material age, stellar lifetime and player session duration are different concepts.

SOURCES S01,S04,S09 / register p.52

Day, year & local orbit

A rotating body and its orbit have separate clocks.

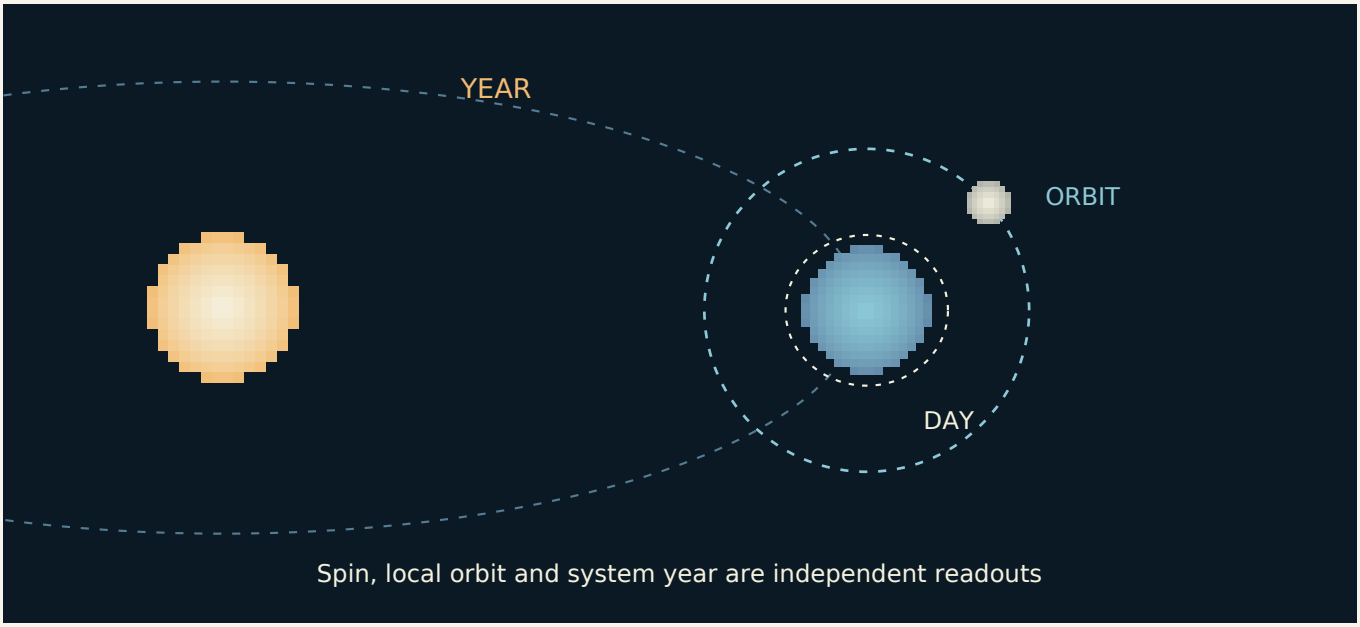


FIG. 44 / Spin period, parent-relative orbit and root-planet stellar year.

$$P_{\text{day}} = \frac{2\pi}{|\omega|}, \qquad \omega = \frac{S}{I} \tag{44.1}$$

$$a_{\text{eq}} = \frac{G(M+m)}{v_{\text{esc}}^2 - v_{\text{rel}}^2}, \qquad P_{\text{eq}} = 2\pi\sqrt{\frac{a_{\text{eq}}^3}{G(M+m)}} \tag{44.2}$$

DAY. The spin clock uses angular velocity derived from stored angular momentum and moment of inertia. A spinless body reports NO SPIN instead of an arbitrarily huge day.

YEAR and ORBIT. A planet's year follows its eligible stellar binding. A moon reports a local parent-relative orbit, while its year follows the root planet's stellar context. Unbound motion does not receive a fabricated periodic year.

Equivalent period. The diagnostic uses the softened escape-energy relation to estimate an equivalent semimajor axis, then a Kepler period. This is useful for a readable HUD, but it is not an exact analytic period for every Plummer-softened trajectory.

Large spans. MYR, GYR and TYR are formatting units for large durations. Their presence is not a prediction that the star will physically live that long or that the simulation resolves stellar evolution on those spans.

DIAGNOSTIC / The equivalent-orbit expression requires a positive binding denominator.

SOURCES S04,S08,S09 / register p.52

The observer's clock

Navigation telemetry is not relativistic matter dynamics.

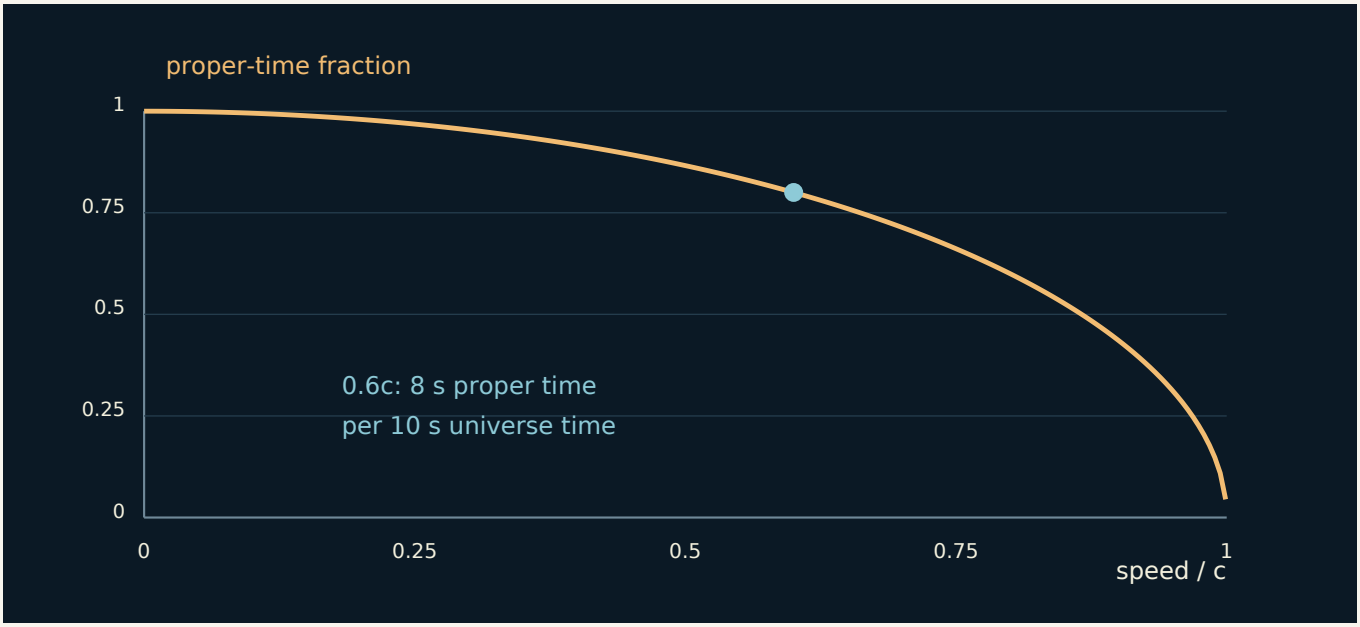


FIG. 45 / Calculated special-relativistic proper-time fraction for a subluminal observer.

$$\beta = \frac{v}{c}, \qquad \gamma = \frac{1}{\sqrt{1-\beta^2}}, \qquad \Delta\tau = \Delta t\sqrt{1-\beta^2} \tag{45.1}$$

$$\Delta t - \Delta\tau = \Delta t\,\frac{\beta^2}{1+\sqrt{1-\beta^2}} \quad (0 \le \beta < 1) \tag{45.2}$$

Subluminal display. The observer's speed relative to the modeled reference frame supplies beta and gamma. Proper-time lag is accumulated using a numerically stable rearrangement, avoiding subtraction of almost equal numbers at low speed.

Example. At 0.6 times light speed, gamma is 1.25. During ten universe seconds the ideal inertial observer accumulates eight seconds of proper time and two seconds of lag.

Explicit boundary. This telemetry does not make matter motion relativistic. It omits gravitational time dilation and light-delayed rendering. LIGHT remains a distance-over-light-speed informational value.

Beyond light speed. Camera navigation may exceed c for usability. Gamma then has no special-relativistic real value and is not displayed as a physical result. The UI's lag policy in that mode is a navigation convention, not a theory of faster-than-light clocks.

CONSTANT / c = 299,792.458 km/s. Freeze navigation has its own nonphysical display policy.

SOURCES S04,S09 / register p.52

Reading the interaction HUD

Ask which mass, which temperature and which reference frame.



FIG. 46 / Annotated schematic HUD, with grouped inventory separated from retained-body state.

Body metrics. CORE, SURF and gas IN/OUT describe different reservoirs. BODY SPEED concerns matter motion. Diameter describes the relevant body or explicitly labelled group, not necessarily the compact giant target.

Group metrics. BODIES greater than one announces an interaction group. GROUP MASS, GROUP DIAM and grouped composition can include a cloud still in transit. They are not proof that its whole inventory has been captured.

Branch metrics. GAS+ and GIANT+ should be interpreted through their retained composition and compressed-size gates. A star label belongs to the gas-only stellar branch, not to an opaque hot planet.

Observer metrics. TIME, PLAY, ODO and LIGHT answer different navigation questions. A large camera speed or odometer says nothing by itself about parcel escape. For diagnosis, record the selected body, its owner, inventory, relative motion and current save/build version.

READING CHECK / First distinguish one body from a group; then compare the appropriate composition and size metrics.

SOURCES S01,S04,S11,S13 / register p.52

Save format & migration

A saved state should use the current model without inventing missing history.

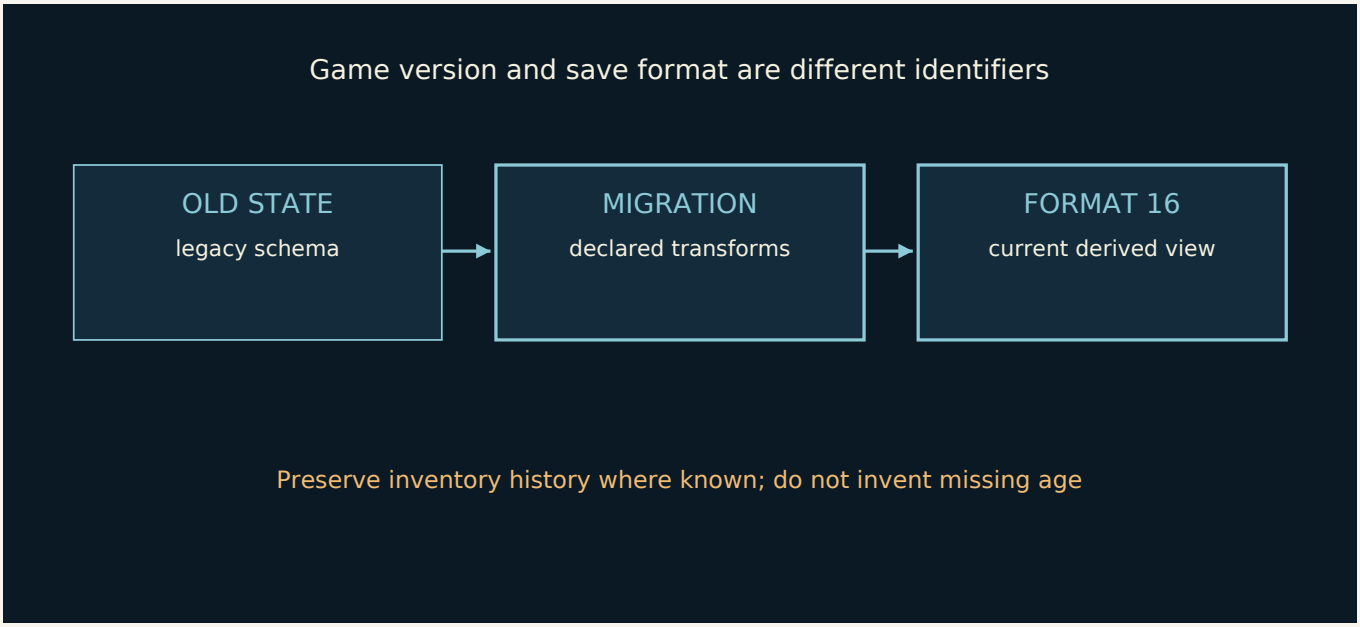


FIG. 47 / Historical saves migrate through declared density and time-schema changes.

$$v_{\mathrm{rel,\,new}} = v_{\mathrm{rel,\,old}} \sqrt{\frac{M_{\mathrm{host,\,new}}}{M_{\mathrm{host,\,old}}}} \tag{47.1}$$

Two version numbers. This atlas covers game 16.3.1-dev and save format 16. A release label and a binary save schema are not interchangeable. A new document edition does not bump either one.

Density migration. The v16 scale transition converts eligible older material inventories to the real-density interpretation while preserving documented geometry and state. Stellar and large-gas exceptions retain their legacy masses; migrations are not a blind rescale of every entity.

Orbital and temporal continuity. The migration adjusts relevant moon-relative velocities with the square root of host-mass change. Missing birth history begins at load, and absent proper-time lag initializes to zero.

Derived state. Current rendering, ownership, zones and HUD values are recomputed from migrated state. The book does not modify any user's saves. A changed palette should not be frozen forever into an old save, but persistent physical state must still be migrated deliberately.

SAFETY / Keep original saves before a schema conversion; this PDF task did not open or rewrite gameplay saves.

SOURCES S01,S02,S04,S05 / register p.52

Runtime architecture

A model is maintainable when its responsibilities are named.

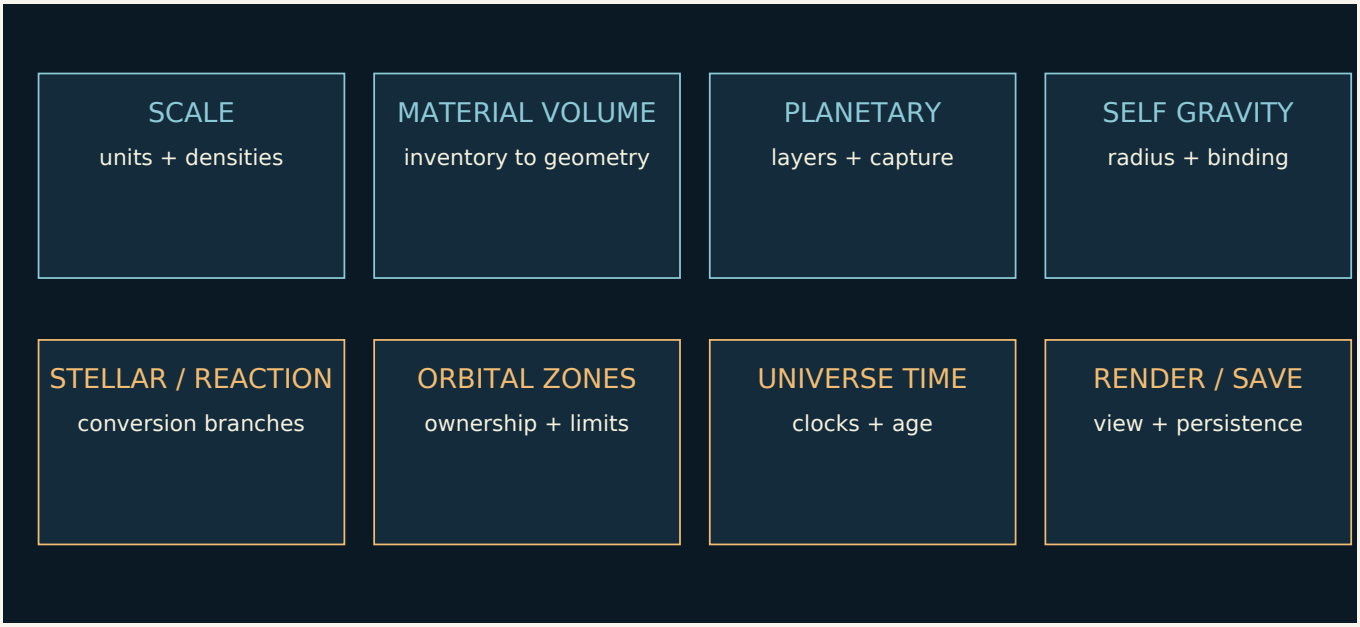


FIG. 48 / Conceptual responsibility map across the current Zig simulation and renderer.

Scale and matter. scale.zig centralizes physical scale, reference density and units. material_volume.zig translates inventory into supported geometry. planetary.zig handles layered planetary state and the gas-solid branch.

Stars and reactions. self_gravity.zig supplies contraction targets and binding heat. stellar.zig, fusion.zig and reactions.zig distinguish stellar status, fuel/product conversion and other material transformations.

Environment and motion. radiation.zig supplies thermal-source behaviour. orbital_zone.zig describes local ownership, Hill/Roche constraints and related orbital decisions. universe_time.zig keeps time and observation semantics explicit.

Presentation and persistence. HUD and render paths read physical state but must not become alternate mass ledgers. Save migration preserves or deliberately reconstructs persistent fields. The file map is a navigation aid; newer modules and handoff notes take precedence over stale descriptions.

MAINTENANCE / This diagram records responsibility boundaries, not every call or file in the repository.

SOURCES S01,S02,S03,S04,S13 / register p.52

Bounded computation

An entity limit is useful only if it does not erase the conservation problem.

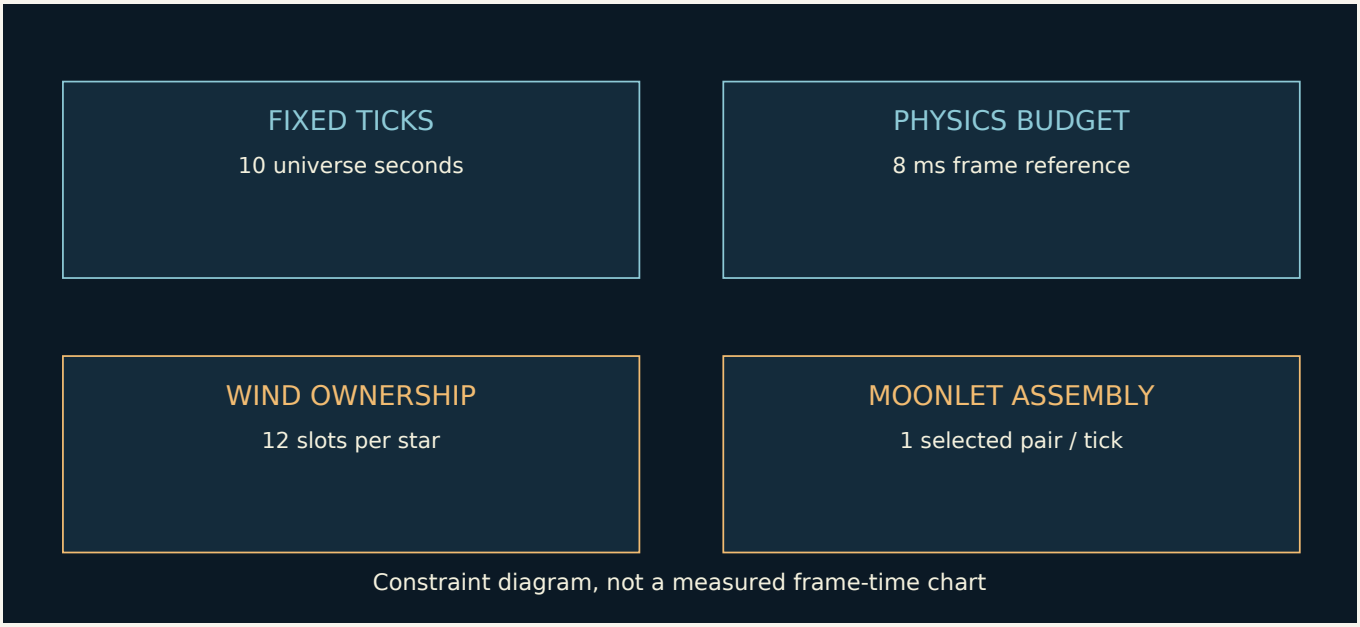


FIG. 49 / A conceptual frame budget with bounded contacts, wind slots and moonlet work.

Finite representation. A universe with a bounded body pool cannot resolve every molecule or dust grain. Breakup must partition real mass into a manageable number of parcels, and rejected spawns must leave that mass somewhere explicit.

Bounded mechanisms. Wind slots limit stellar packet ownership. Contact batches limit repeated collision work. Moonlet assembly handles at most one chosen pair per tick. Time warp stays inside the frame scheduler's physics budget.

Different costs. Pair searches, swept collision tests, thermal exchange and pixel glow rendering stress different parts of the runtime. Reducing the visible number of dots is not evidence that GPU time or collision cost improved.

Open case. Multi-parcel Roche overflow can still crowd a scene. The proposed aggregate ring is a representation change that needs its own ledger and acceptance tests. This atlas reports prior results; it does not contain a new GPU benchmark or measured frame-time guarantee.

PERFORMANCE STANDARD / Report scene, body count, simulation rate, CPU/GPU timing method and high-percentile frame times.

SOURCES S01,S03,S04 / register p.52

What the evidence establishes

Recorded checks are useful, but they are not a new gameplay acceptance run.



FIG. 50 / Reported numerical coverage versus still-unrecorded v16 gameplay acceptance.

Recorded baseline. The supplied October-7 documentation reports build 3/3, physics 403/403, save 447/447, rendering 517/517 and build-test orchestration 17/17 steps. universe_time.md reports 1,260/1,260 tests for its recorded build run.

Scale audit. The handoff reports zero failing targets out of thirteen scale-audit targets, while explicitly retaining twenty-three absolute mechanical proxies. Passing the audit does not mean every rule became first-principles physics.

Evidence boundary. These counts come from project records and may overlap; they are not additive independent samples. No game tests or builds were executed to produce this book. The handoff does not record v16.1-v16.3.1 visual gameplay acceptance.

Document checks. PDF generation separately checks page count, equation placement, text bounds, metadata and rendered layout. Those checks establish the quality of this artifact, not the correctness of every simulation scenario shown schematically.

REPORTED, NOT RERUN / Keep numerical regressions, performance measurements and user visual acceptance as separate evidence.

SOURCES S01,S02,S04,S05 / register p.52

Limits & the next questions

A faithful atlas shows where the model stops.



FIG. 51 / Current mechanics, declared approximations and explicitly proposed development.

Current foundations. Real reference density and gravity, mass-paid parcel contact, separate thermal reservoirs, compressed-volume geometry and bounded orbital zones now share a clearer scale. That is progress without claiming a complete astrophysical solver.

Declared reductions. Cold reference densities, prescribed gas compression, Plummer softening, accelerated radiation, fixed layer fractions and fitted mass-radius curves remain approximations. The rendered universe is planar even when its packet mass uses spherical volume.

Still proposed. Aggregate rings, richer transport inside annuli and further moon/ring evolution require new implementation and save decisions. Neither a diagram nor a future branch name counts as completed support.

Next acceptance questions. Can a repeatable giant recipe settle without free kicks? Do hot gas wakes remain visible at useful zoom? Does a ring-forming overflow case stay bounded in cost? Do migrated saves preserve intended inventories and show current rendering? Record the answers with versioned scenes.

DEVELOPMENT ETHIC / Preserve the working model, distinguish intention from evidence and revise one contract at a time.

SOURCES S01-S05 / register p.52

Sources & equation index

Local implementation records are the authority for this edition.

S01 **AGENT_HANDOFF_v16.3.1.md**
Supplied handoff; implementation status and recorded evidence.

S02 **docs/SCALE_CONGRUENCE.md**
Density normalization, scale audit and remaining proxies.

S03 **docs/PLANETARY_ORBITAL_ZONES.md**
Ownership, zones, Roche overflow, belts and roadmap.

S04 **docs/UNIVERSE_TIME.md**
Universe clocks, warp, ages, orbit readouts and telemetry.

S05 **CHANGELOG.md; VERSION**
Version history and current development label.

S06 **src/scale.zig**
Physical constants, density references and unit bridge.

S07 **src/self_gravity.zig**
Mass-radius targets, contraction and binding heat.

S08 **src/orbital_zone.zig**
Hill/Roche geometry and derived orbital decisions.

S09 **src/universe_time.zig**
Fixed-step time and observer-clock functions.

S10 **src/material_volume.zig**
Inventory-volume conversion and compression reference.

S11 **src/planetary.zig**
Layered planet constants and gas-growth thresholds.

S12 **src/stellar.zig; fusion.zig; reactions.zig**
Stellar gates and material-conversion branch constants.

S13 **docs/FILE_MAP.md + current handoff**
Runtime responsibility and HUD context; stale entries qualified.

S14 **src/radiation.zig**
Thermal source and rendering-related reference parameters.

H01 **output/pdf/UNIVERSAL_TOPOLOGY_V3.pdf**
43-page v15 historical atlas; preserved, not overwritten.

EQUATION LOCATOR

Mass and projection: 6-11

Units and gravity: 13-14

Momentum and contact: 15-18

Heat and composition: 19-22

Giant and stellar branches: 23-30

Orbital assembly: 32-40

Universe and observer clocks: 42-45

Migration velocity scaling: 47

Original concept and project material: **GitHub @kujiwa-rgb / X @officialemane**. Copyright 2026. Standard scientific formulas are not claimed as original inventions. This edition documents a local implementation snapshot, not a complete astrophysics solver.

Historical numerical examples are superseded only where this edition states the newer rule.